Cycle financing and a period lower bound for the Juggler map

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Abstract

The Juggler map is the nonlinear integer map

$$J(n) = \begin{cases} \lfloor \sqrt{n} \rfloor, & n \text{ even}, \\ \lfloor n^{3/2} \rfloor, & n \text{ odd}. \end{cases}$$

It is conjectured that every positive integer eventually reaches 1.

We develop a cycle-financing inequality for this floor-power map. Exact integer cells give a one-step logarithmic defect; cycle minimality lets that defect be unrolled against the cycle minimum; the formal surplus $3^o - 2^L$ must then be paid by a finite accumulated budget. For a hypothetical cycle of length L with o odd steps and minimum n,

$$n \log n \cdot (3^o - 2^L) \leq L \cdot 3^o.$$

As a consequence, combining the inequality with the known verification through 10^6 yields $L \geq 25781$; at the laboratory-certified descent floor $N_0 = 26254995$ the same table yields $L \geq 50508$. The novelty is not a new computational record; it is that implication. A census-free walk-charge envelope — transport of the floor losses to a reduced base, identification of the extremal exponent walk as a rotation word, and a Denjoy–Koksma bound over certified Ostrowski blocks — then extends the exclusion at the laboratory floor: any nontrivial cycle has period at least 176251; a second certified floor, $N_0 = 162849448$, raises the bound to 478245. We also prove that every nontrivial cycle contains at least four even steps, and hence has period at least eleven; that bound uses no descent floor. The core arguments are formalized in Lean 4; the finite numerical tables are independently certified computations.

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1. Introduction

The Juggler sequence was introduced by Pickover [1,2]. The one-step map is OEIS A094683 [3]; the stopping-time table is A007320 [4]. The orbit of 3 is $3, 5, 11, 36, 6, 2, 1$. The orbit of 37 already peaks at 24906114455136. Universal arrival at 1 remains the open conjecture of the subject.

The map combines a contracting even branch with an expanding odd branch,

$$E(n) = \lfloor n^{1/2} \rfloor, \qquad O(n) = \lfloor n^{3/2} \rfloor.$$

A word of length k with o odd letters has ideal exponent $3^o/2^k$. Floors are applied after every letter, and a word is available only when the orbit realizes those parities.

The mechanism is the interaction of three elementary facts: exact integer cells, a logarithmic defect, and cycle minimality. Exact cells give a one-step logarithmic defect; cycle minimality lets the defect be unrolled against the cycle minimum; the formal surplus $3^o - 2^L$ must then be paid by a finite accumulated budget. That is Theorem 4.4:

$$n \log n \cdot (3^o - 2^L) \leq L \cdot 3^o.$$

The inequality $\log(1+u) \le u$ is the only analytic input; it is not the content of the theorem.

Write N_0 for a *verified descent floor*: every start $2 \le n \le N_0$ reaches 1. A floor is an input, not the result. Two instances are used. The base instance is $N_0 = 10^6$, reported by Weisstein [5] and recomputed here by exact first-passage. Combined with Theorem 4.4 it gives

$$\text{known verification through } 10^6 \;+\; \text{this inequality} \quad \Rightarrow \quad L \ge 25781.$$

The laboratory instance is $N_0 = 26254995$, certified by the same exact first-passage method (Proposition 5.1); the same table then gives $L \ge 50508$ (Theorem 5.2), and the walk-charge envelope of Section 5 amplifies it to $L \ge 176251$ (Theorem 5.9); the second certified floor $N_0 = 162849448$ raises it to $L \ge 478245$ (Corollary 5.10). The architecture is

$$\text{envelope} \to \text{cycle minimum} \to \text{finance} \to n_{\max}(L) \to N_0 \to L \ge 25781 \text{ at } 10^6,$$

extended at the laboratory floor by

$$\text{transport} \to \text{hug adversary} \to \text{word identity} \to \text{Denjoy–Koksma} \to \text{window} \to L \ge 176251.$$

The computation supplies the endpoint N_0. The mathematics amplifies that floor to the period bound.

Roadmap. Section 2 records the power envelope and the exact defect identity that explains it. Section 3 classifies minimum-based cycle words and proves that every nontrivial cycle has at least four even letters. Section 4 records the excursion necklace of a minimum-based word, unrolls the cell logarithm around that minimum, obtains the finance inequality, and applies it at the known floor 10^6. The necklace is the geometry of the unroll, not a fourth main theorem. A run-type packing of the same identity is a supporting refinement; the arithmetic of the leftover lengths is secondary. Section 5 certifies the laboratory floor 26254995, replaces the length-only charge by a coupled exponent-walk charge, identifies its extremal word as a rotation word, and proves a census-free envelope for every length in the window $[50508, 301994)$; the resulting kill table gives the period bound 176251, and a second certified floor raises it to 478245 (Corollary 5.10). Section 6 records limitations.

Throughout, $\mathbb{N} = \{1, 2, 3, ...\}$. Write J^k for the k-fold iterate. A nonempty realized word w with $J^{|w|}(n) = n$ is a *cycle word*. The unique fixed point is 1; a cycle is *nontrivial* when it contains some $n \ge 2$. A cycle word at n is *minimum-based* when n is a cycle minimum: $J^j(n) \ge n$ for every $0 \le j < |w|$. Every cycle word has a minimum-based rotation.

All exceptional sets below are *finance-survivor* sets: lengths that a stated form of the finance inequality does not exclude at the verified descent floor. They are not candidate cycle sets.

1.0 Main results

Contribution 1 — cycle-financing inequality. For a minimum-based cycle of length L with o odd steps,

$$n \log n \cdot (3^o - 2^L) \le L \cdot 3^o$$

(Theorem 4.4). Floor defects become a quantitative bound on the cycle minimum.

Contribution 2 — structural word obstruction. Every nontrivial cycle word has at least four even letters, and hence period at least eleven (Theorem 3.22 and Corollary 3.23). The argument classifies the minimum-based word geometry; it is not a raw census of words of length at most ten. This bound does not use the verified descent floor.

Contribution 3 — explicit global consequence. Combined with the independently verified descent floor $N_0 = 10^6$, there is no nontrivial Juggler cycle of length at most 25780. Equivalently, any nontrivial cycle has period at least 25781 (Theorem 4.6(A)). At the laboratory-certified floor $N_0 = 26254995$ the same table gives period at least 50508 (Theorem 5.2).

Contribution 4 — census-free walk-charge envelope. On a minimum-based cycle every state is coupled through one closed exponent walk. Transport of the floor losses to a reduced base (Theorem 5.3), identification of the extremal walk as the rotation (hug) word (Theorem 5.4, Lemma 5.6), and a Denjoy–Koksma bound over certified Ostrowski blocks (Theorem 5.7) give a uniform envelope for every length in the window $[50508, 301994)$ (Theorem 5.8). At the laboratory floor the resulting kill table leaves a single finance survivor below $2 \cdot 10^5$: any nontrivial cycle has period at least 176251 (Theorem 5.9). At the second certified floor $N_0 = 162849448$ the same envelope leaves only the semiconvergent fan member 478245: any nontrivial cycle has period at least 478245 (Corollary 5.10).

These statements are not interchangeable. Theorem 4.4 is the conceptual sharp inequality (constant 1). Corollary 4.5 is the convenient length-only statewise bound that turns a descent floor into a per-length exclusion. Theorem 4.6 is the numerical certification of that bound, and it uses a conservative coefficient $6/5$ only as a uniform majorant; the headline cutoff 25781 is not an artifact of that majorant. Theorem 4.7 is a run-type refinement of the same defect sum. The leftover lengths through 10^5 are supporting material, not a second main theorem.

1.1 Related work

Pickover's later exposition is Chapter 45 of [2]. Weisstein [5] records the map, the stopping-time sequences A094670, A094679, A095908, and a verification of arrival at 1 through 10^6. That verified descent floor is the computational input to Theorem 4.6; the first-passage run of Appendix B recomputes it. OEIS A094716 [6] records extreme heights, including the start 48443 whose peak has 972 463 digits. Those height records do not bound the period.

Prasad–Prasad [7] estimate excursion and stopping constants for juggler-like maps by a random-walk large-deviation model; those estimates do not apply to exact cycles. Small-cycle censuses are a standard first layer for Collatz-like maps, surveyed by Lagarias [8,9]. Those results do not transfer: the branches of J are floor powers rather than affine maps (Crandall [10], Matthews–Watts [11]). In particular there is no identity of the form $n(2^K - 3^p) = C$. A check of Pickover [1,2], Weisstein [5], the OEIS records [3,4,6], the Prasad–Prasad estimates [7], and the standard Collatz cycle-bound sources [8–13] found no published explicit lower bound on the period of a nontrivial cycle for this exact floor-power Juggler map. Informal near-miss notes and later unverified webpage floors are not used.

The layers of the argument are as follows.

1. *Classical idea:* cycle financing (Simons–de Weger [12]).
2. *Classical idea:* logarithmic and continued-fraction approximation of $\log 2/\log 3$.
3. *Classical idea:* cycle-word restrictions and leftover packaging (Eliahou [13]; Lagarias [8,9]).
4. *New object:* the exact one-step floor-power cells of J.
5. *New theorem:* the Juggler-specific cycle-minimum finance inequality of Theorem 4.4.
6. *New consequence:* $L \geq 25781$ at the verified descent floor 10^6.
7. *Supporting organization:* the run-packing refinement of the same defect sum (Theorem 4.7).
8. *Classical ideas:* the Denjoy–Koksma inequality and Ostrowski numeration, used as known tools in Section 5.
9. *New theorem:* the reduced-base transport and the uniform window envelope of Section 5 (Theorems 5.3 and 5.8).
10. *New consequence:* period at least 176251 at the laboratory floor 26254995 (Theorem 5.9), and 478245 at the second certified floor 162849448 (Corollary 5.10).

Financing as an idea, logarithmic step inequalities, the continued-fraction phenomenon, the Denjoy–Koksma inequality, and Ostrowski numeration are not claimed as new. The argument below is elementary and independent of the Diophantine tools of [12]: floor-power defects are relatively $O(1/x)$ in logarithms, so a uniform logarithmic floor-error bound, valid above the verified floor, excludes every length that is not a finance-survivor for Theorem 4.4. To the best of our knowledge, the Juggler-specific inequality and the explicit period bound it produces are new.

Novelty statement. For this nonlinear floor-power map, exact floor defects convert into a cycle-financing inequality that forces the minimum of a hypothetical cycle below the independently verified descent region unless the period is at least 25781. Coupling the states through the closed exponent walk and bounding the extremal rotation word by Denjoy–Koksma over certified Ostrowski blocks raises that period bound to 176251 at the laboratory floor, census-free on an explicit window of lengths, and to 478245 at the second certified floor.

1.2 Verification

The arguments of Sections 2, 3, and 4 may be read without machine assistance. The core mathematical lemmas are mechanized in Lean 4; selected finite classifications and numerical tables are independently certified computations. Appendix A records the Lean names. The finite tables used by Lemmas 3.5, 3.7, 3.11 and Theorems 3.12–3.20 are `native_decide` evaluations in the modules named there.

Proposition 1.3 (certified computational input). A machine-verifiable certificate establishes that every integer $2 \le n \le 10^6$ reaches 1. Precisely: an exact-integer first-passage run records, for each such n, a finite realized word with image strictly below the start; strong induction on that image reaches 1. The longest first passage in the window has 253 steps (seed 78901). Weisstein [5] records the same computational verification; the run here is an independent recomputation. The certificate files, SHA-256 hashes, and regeneration commands are Appendix B.

Roles. Independently proved: the finance inequality (Theorem 4.4). Computational input: every $2 \le n \le 10^6$ reaches 1 (this proposition), every $2 \le n \le 26254995$ reaches 1 (Proposition 5.1, the laboratory instance), and every $2 \le n \le 162849448$ reaches 1 (Corollary 5.10, the second laboratory instance). Independently recomputed: the exact first-passage runs of Appendix B. Not proved: global termination.

Theorem 4.6 applies Corollary 4.5 to this input and certifies the table with the conservative coefficient 6/5; its certified identity is Lean (`cycleMin_defect_finance`, `DefectFinance.lean`), the per-length table stays a verified computation. Theorem 4.7 is a human proof; Theorem 4.8 reuses the gap table under that packing. Proposition 4.9 is integer arithmetic in Lean. In Section 5, Theorem 5.7 is a human proof (Denjoy–Koksma is used as a known tool; its per-block hypotheses — convergent quality $|\theta - p/q| < 1/q^2$ and the block permutation — are Lean, `theta_convergent_quality`, `theta_block_permutations`) and so is the rotation identification in Lemma 5.6, whose word identity itself is Lean (`budgetedWord_eq_hugWord`); the Laplace bound of Proposition 5.5 is Lean (`rotation_average_le`, `rotationAverage_gap`; the ergodic identification stays human); the transport inequality of Theorem 5.3 (`cycleMin_transport`), the defect-to-hug-charge consequence of §5.2 (`cycleMin_defect_le_charge`, `cycleMin_defect_le_hug_charge`), the charge-maximisation half of Theorem 5.4 (`hug_charge_maximal`; strict uniqueness stays human), the digit-cap step and digit scan of Theorem 5.8 (`ostroDigit_le`, `theta_digitSum_le`, `window_digit_scan`), and the kill template of Theorem 5.9 (`cycleMin_hug_kill_criterion`) are Lean-verified; Theorems 5.2 and 5.9 are independently certified computations at the laboratory floor, with the same trust boundary as Theorem 4.6 (exact integer arithmetic plus guarded float comparisons). This is not a claim that the paper as a whole is formally verified.

```
Repository:  https://github.com/sneakyweasel/balanced_ternary
Commit:      7802f78bec58c68cec92a2efb3db4a502f916277
Lean:        leanprover/lean4:v4.33.0
Mathlib:     v4.33.0 (lake-manifest rev db584cd6d46c92f209a44c0f1c829460d327499d)
Build:       lake build Problems.JugglerPaper   (from formal/)
Computation: python -m research.juggler_sequence.cycle_finance
             (parity table); run-type table in budget_opt.json
SHA-256:     Appendix B
```

The commit is the repository state that produced the finance tables. A later editorial commit of this text does not change those hashes.

2. Envelope

Let $\mathcal{B} = \{E, O\}$. A finite word $w \in \mathcal{B}^*$ is *realized* at $n \in \mathbb{N}$ when the successive parities of the orbit of n are exactly the letters of w. Write $J^{|w|}(n)$ for the endpoint after those letters, and $\#O(w)$ for the number of odd letters. When w is realized, this endpoint is the $|w|$-fold iterate.

Theorem 2.1 (fixed-word monotonicity). If $n \le m$ and both realize w, then $J^{|w|}(n) \le J^{|w|}(m)$.

Proof. Induct on w. The empty word is immediate. The images after a common realized prefix remain ordered, and both current states realize the same next letter. The even branch is the monotone integer square root; the odd branch is $x \mapsto x^3$ followed by the integer square root. □

The realizing set of a fixed word need not be an interval: OE is realized at 7 and at 11 but not at 9.

Theorem 2.2 (finite-word power envelope). If w is realized at n and $m = J^{|w|}(n)$, then

$$m^{2^{|w|}} \le n^{3^{\#O(w)}}.$$

Proof. The empty word is the equality $n \le n$. Suppose a realized prefix of length ℓ with odd count o ends at x and satisfies $x^{2^\ell} \le n^{3^o}$, and the next letter is realized.

If the next letter is even, then $J(x)^2 \le x$. Raising the inductive bound to the second power gives

$$J(x)^{2^{\ell+1}} = (J(x)^2)^{2^\ell} \le x^{2^\ell} \le n^{3^o}.$$

The odd count is unchanged.

If the next letter is odd, then $J(x)^2 \le x^3$, so

$$J(x)^{2^{\ell+1}} = (J(x)^2)^{2^\ell} \le x^{3\cdot 2^\ell} = (x^{2^\ell})^3 \le \left(n^{3^o}\right)^3 = n^{3^{o+1}}.$$

This is the claimed bound after one more odd letter. □

Corollary 2.3 (exponent-gap contraction). If $n \ge 2$, w is realized at n, and $3^{\#O(w)} < 2^{|w|}$, then $J^{|w|}(n) < n$.

Proof. Let $m = J^{|w|}(n)$ and $k = |w|$. Theorem 2.2 gives $m^{2^k} \le n^{3^{\#O(w)}}$. The exponent gap and $n \ge 2$ give $n^{3^{\#O(w)}} < n^{2^k}$, so $m^{2^k} < n^{2^k}$. Since $m \ge 1$, one has $m < n$. □

The corollary includes familiar contracting blocks such as $OOOEE$. It does not prove that every start realizes some contracting word.

2.4 Exact floor defect

Theorem 2.2 is the inequality form of an exact identity

$$n^{3^{\#O(w)}} = J^{|w|}(n)^{2^{|w|}} + \Delta_w(n), \qquad \Delta_w(n) \ge 0.$$

The identity, its vanishing law, and the two-term composition are Theorems 2.4–2.6 and Corollary 2.7 in Appendix C. A mixed realized word has $\Delta_w(n) > 0$. This explains why the envelope of Theorem 2.2 is true, and why a mixed cycle word is formally expanding. Theorem 4.4 uses only the nonnegativity $\Delta_w(n) \ge 0$.

No uniform local tax exists: the relative slack of a single letter tends to 0 with the state. Finance must therefore be a global comparison, not a fixed cost per odd or even step. A quantitative itinerary-dependent lower bound on Δ_w, or a tighter upper bound on $\sum 1/(x_i \log x_i)$, would improve the period cutoff; neither is proved here.

3. Structural restrictions on cycle words

The headline of this section is Theorem 3.22: every nontrivial cycle word has at least four even letters, and hence period at least eleven. The small-period censuses (Theorems 3.6 and 3.8) are supporting. After the one-step cells, a minimum-based word has a canonical run form. There are only three even-count regimes with $e \le 3$; each reduces to a finite collection of geometries $O^aEO^bEO^cE$. Those geometries are eliminated by a next-square obstruction, by long odd-run growth against a last-even cell, or by a finite exceptional window. The family calculations are Appendix D.

The exact one-step fibers are

$$J(n) = q \iff q^2 \le n < (q+1)^2 \quad (n \text{ even})$$

and

$$J(n) = m \iff m^2 \le n^3 < (m+1)^2 \quad (n \text{ odd}).$$

Lemma 3.1 (odd cells are unique). An odd fiber contains at most one integer. An even fiber is a parity-restricted square interval and may contain many predecessors.

Proof. Suppose $a < b$ lie in the same odd cell indexed by m. Then $(a+1)^3 \le b^3 < (m+1)^2$ and $m^2 \le a^3$. Subtracting the latter lower bound from the former upper bound gives

$$3a^2 + 3a + 1 = (a+1)^3 - a^3 < (m+1)^2 - m^2 = 2m + 1,$$

hence $3a^2 + 3a < 2m$ and $3a^2 < 2m$. If $a = 0$, then $m = 0$ and the displayed strict inequality is already impossible. If $a > 0$, squaring and using $m^2 \le a^3$ gives $9a^4 < 4m^2 \le 4a^3$, contradicting $4a^3 < 9a^4$. □

Theorem 3.2 (cycle restrictions). Let w be a cycle word at $n \ge 2$.

(i) The word is formally expanding:
$$2^{|w|} < 3^{\#O(w)}.$$
A contracting word cannot close a nontrivial cycle.

(ii) The cycle minimum is odd and the cycle maximum is even. A minimum-based orientation cannot end in an odd letter.

(iii) A realized word v is *superquadratic* if
$$3^{\#O(v)} \ge 2^{|v|+1}.$$
Any realized path from a start $n \ge 2$ to a state at least n^2 is superquadratic. On a cycle minimum the path to any later even state is superquadratic. The prefix *OOE* is expanding ($9 > 8$) but not superquadratic ($9 < 16$), so it cannot carry the minimum to square scale.

Proof. For (i), Theorem 2.2 applied to a cycle endpoint gives $n^{2^{|w|}} \le n^{3^{\#O(w)}}$. Since $n \ge 2$, comparison of exponents gives $2^{|w|} \le 3^{\#O(w)}$; equality is impossible because the two sides have different prime divisors and $|w| \ge 1$. Alternatively: a mixed cycle word has $\Delta_w(n) > 0$ by Theorem 2.5 in Appendix C, so the envelope is strict; a monochrome tower cannot return for $n \ge 2$.

Every state on such a cycle is at least 2: once an orbit reaches 1, it remains there and cannot return to a start $n \ge 2$. An even state $x \ge 2$ satisfies $J(x) < x$, so a cycle minimum cannot be even. An odd state $x \ge 3$ satisfies $J(x) > x$, so a cycle maximum cannot be odd. This proves the first assertion of (ii). For the final-letter assertion, let x be the predecessor of a minimum-oriented return n. If the last letter were odd, the odd return cell would give $n^2 \le x^3 < (n+1)^2$. Minimality gives $x \ge n$, hence $n^3 < (n+1)^2$, impossible for $n \ge 3$; and the minimum is odd, so $n \ge 3$.

For (iii), let a realized word v send n to $y \ge n^2$. Theorem 2.2 gives

$$n^{2^{|v|+1}} = (n^2)^{2^{|v|}} \le y^{2^{|v|}} \le n^{3^{\#O(v)}}.$$

Since $n \ge 2$, comparison of exponents proves the superquadratic inequality. On a cycle minimum, if a later state y is even, its successor is at least n. Thus $n \le J(y) = \lfloor \sqrt{y} \rfloor$, so $n^2 \le y$, and the preceding argument applies to that prefix. □

Lemma 3.21b (canonical run form). After rotation to a minimum-based orientation, the word begins with *OO* and ends with E; hence every word with $1 \le e \le 3$ even letters has the canonical run decomposition $O^aEO^bEO^cE$ (unused runs empty). The case $e = 0$ is all-odd and is already forbidden by the last-letter restriction. No other cyclic rotation needs a separate case.

Proof. Theorem 3.2: the minimum is odd, so the word cannot begin with E or *OE*, and it cannot end with an odd letter. Thus a minimum-based word starts *OO* and ends E, so $e \ge 1$. The remaining letters are odd runs separated by the e even letters, so the word is $O^{a_1}E\cdots O^{a_e}E$ with $a_1\geq 2$. For $e\leq 3$ this is $O^aEO^bEO^cE$ with unused runs empty. Every cycle word has a minimum-based rotation of the same even-count, and that orientation is already in this form. □

Thus it is enough to exclude the three regimes $e=1,2,3$. The rest of the section does that. Theorems 3.6 and 3.8 record the short-period consequences; Theorem 3.22 is the structural statement.

Lemma 3.3 (coarse lower envelope). For $n\geq 1$, write $q=\lfloor\sqrt{n}\rfloor$. Then $q^2\leq n<(q+1)^2$. For $q\geq 1$ one has $(q+1)^2\leq 4q^2$, and the second inequality is strict for $q\geq 2$, while for $q=1$ one has $n<4$. In all cases

$$n<4\lfloor\sqrt{n}\rfloor^2.$$

Thus an even step satisfies $n\leq 4\,J(n)^2$ and an odd step satisfies $n^3\leq 4\,J(n)^2$. Composing along a realized word v of length k with odd count o gives

$$n^{3^o}\leq C_v\,J^k(n)^{2^k}.$$

The constant starts at 1 and updates by $C\mapsto C\cdot 4^{2^j}$ on an even letter and $C\mapsto C^3\cdot 4^{2^j}$ on an odd letter, at step j. In particular $C_{OOO}=2^{38}$ and $C_{OOOO}=2^{130}$.

Proof. The comparison $n<4q^2$ is the paragraph above. The composition law is the same recurrence as the proof of Theorem 2.2, with a factor 4^{2^j} inserted at each letter. The values C_{OOO} and C_{OOOO} are the result of that recurrence on those two words. □

Lemma 3.4 (next-square thresholds). (i) If $q\geq 5$ realizes OO, then $J^2(q)\geq(q+1)^2$. (ii) If $q\geq 3$ realizes OOO, then $J^3(q)\geq(q+1)^2$. (iii) If a realized word v satisfies $J^{|v|}(q)\geq(q+1)^2$ and the next realized letter is odd, then $J^{|v|+1}(q)\geq(q+1)^2$. (iv) If vE is a cycle word at n, then $J^{|v|}(n)<(n+1)^2$. (v) If $a\geq 3$, then O^aE is not a cycle word at any $n\geq 2$.

Proof. For (i), write $m=\lfloor q^{3/2}\rfloor$. The bound $J^2(q)\geq(q+1)^2$ follows from $m^3\geq(q+1)^4$, because then $\lfloor m^{3/2}\rfloor\geq(q+1)^2$. If $q=5$, then $11^2=121\leq 125=5^3<144=12^2$, so $m=11$, and $11^3=1331\geq 6^4=1296$. Since q realizes OO, it is odd, so the only remaining case is $q\geq 7$. Then $m\geq q^{3/2}-1$, so it is enough that $(q^{3/2}-1)^3\geq(q+1)^4$. Expanding the left side and dropping the positive remainder $3q^{3/2}-1$ reduces this to $q^{9/2}-3q^3\geq(q+1)^4$, or equivalently $\sqrt{q}-3/q\geq(1+1/q)^4$. The left side increases for $q>0$ and the right side decreases, so the case $q=7$ suffices:

$$\sqrt{7}-\frac{3}{7}>\frac{5}{2}-\frac{3}{7}=\frac{29}{14}>\frac{4096}{2401}=\left(\frac{8}{7}\right)^4,$$

where $\sqrt{7}>5/2$ follows from $25/4<7$.

For (ii), the orbit $3\to 5\to 11\to 36$ gives $J^3(3)=36\geq 16$. If $q\geq 5$ realizes OOO, then it realizes OO, so (i) gives $J^2(q)\geq(q+1)^2$. The third letter is odd, hence $J^3(q)=J(J^2(q))\geq J^2(q)$.

For (iii), the image after v is odd and at least $(q+1)^2\geq 4$, so the odd branch does not decrease it.

For (iv), the last letter is even, so the preimage $z=J^{|v|}(n)$ is even and satisfies $n^2\leq z<(n+1)^2$.

For (v), suppose O^aE is a cycle word at $n\geq 2$. The start is odd, hence at least 3, and realizes O^a. Parts (ii) and (iii) give $J^a(n)\geq(n+1)^2$, contradicting (iv). □

Lemma 3.5 (two length-six exclusions). Neither $OOOEOE$ nor $OOOOEE$ is a cycle word at any $n\geq 2$.

Proof. First, if $n\geq 256$, then

$$n^{81}>2^{130}(n+1)^{64}.$$

Indeed $257^{64}<2\cdot 256^{64}$ because $(1+1/256)^{64}<e^{64/256}=e^{1/4}<2$, and $256(n+1)\leq 257\,n$, so $(n+1)^{64}<2n^{64}$. Then $2^{130}(n+1)^{64}<2^{131}n^{64}$. Since $n\geq 256=2^8$, one has $n^{17}\geq 2^{136}$, and therefore $2^{131}n^{64}<2^{136}n^{64}\leq n^{81}$.

For $2\leq n<256$, neither word returns to its start. This is a table of 254 evaluations of each word: at every start, either some letter fails to match the current parity, or the six-step image differs from the start. The same finite check is the Lean `native_decide` evaluation behind `no_cycle_word_oooeoe` and `no_cycle_word_oooooee` (Appendix A).

Now suppose $OOOOEE$ is a cycle word at $n \geq 256$, and write $z = J^4(n)$ for the image after the prefix $OOOO$. Lemma 3.4(iv) gives $J(z) < (n+1)^2$, and the preceding even letter gives $z < (J(z)+1)^2$, hence $z < (n+1)^4$. Lemma 3.3 on $OOOO$ gives $n^{81} \leq 2^{130} z^{16}$, so $n^{81} < 2^{130}(n+1)^{64}$, contradicting the tail inequality.

Finally suppose $OOOEOE$ is a cycle word at $n \geq 256$. Write $z_3 = J^3(n)$ and $y = J(z_3) = \lfloor \sqrt{z_3} \rfloor$, so $z_3 < (y+1)^2$. Lemma 3.3 on OOO gives $n^{27} \leq 2^{38} z_3^8 < 2^{38}(y+1)^{16}$. Cubing yields $n^{81} < 2^{114}(y+1)^{48}$. The last letters OE give the odd-cell bound $y^3 < (n+1)^4$. Write $A = n+1 \geq 257$. We claim $(y+1)^3 < 2A^4$. If $y \leq A$, this is $(A+1)^3 < 2A^4$. If $y > A$, then $y \geq 258$, so $3y+1 < y^2$ and hence $(y+1)^3 = y^3+3y^2+3y+1 < A^4 + 4y^2$, while $4y^2 < A^4$ because $4y^3 < 4A^4 \leq yA^4$. Raising $(y+1)^3 < 2A^4$ to the sixteenth power gives $(y+1)^{48} < 2^{16}(n+1)^{64}$. Combining with the cubed lower envelope produces again $n^{81} < 2^{130}(n+1)^{64}$. □

Theorem 3.6 (small-cycle census). No word of length at most six is a cycle word at any $n \geq 2$. Equivalently, a nontrivial Juggler cycle, if one exists, has period at least seven.

Proof. Rotating a cycle word by one letter moves the base point one step along the orbit and yields another cycle word; every state of the cycle is at least 2, because an orbit that reaches 1 stays at 1 and cannot return to a start $n \geq 2$.

If every letter is odd, the start is odd, hence $n \geq 3$, and the odd branch strictly increases there: $J(x) > x$ for odd $x \geq 3$, since $x^3 \geq (x+1)^2$. The orbit ascends strictly and never returns. Otherwise some letter is even, and a rotation ending just after that letter produces an even-terminating cycle word vE of the same length based at a cycle state $m \geq 2$. It therefore suffices to exclude even-terminating cycle words of length at most six.

By Theorem 3.2(i) a cycle word is formally expanding. No even-terminating word of length one or two is expanding ($2 > 3^0$ and $4 > 3$).

Length three: the only expanding candidate is OOE. If $m \geq 5$ realizes OO, Lemma 3.4(i) gives $J^2(m) \geq (m+1)^2$, contradicting Lemma 3.4(iv). The only smaller odd start is $m = 3$, where $J^2(3) = 11$ is odd, so the final even letter is not realized.

Length four: the expanding filter requires three odd letters among the first three, leaving only O^3E, excluded by Lemma 3.4(v).

Length five: the filter requires four odd letters among the first four, leaving only O^4E, excluded by Lemma 3.4(v).

Length six: the filter requires at least four odd letters among the first five, leaving O^5E, $EOOOOE$, $OEOOOE$, $OOEOOE$, $OOOEOE$, and $OOOOEE$. Lemma 3.4(v) excludes O^5E. The word $EOOOOE$ rotates one step onto $OOOOEE$, and $OEOOOE$ rotates two steps onto $OOOEOE$; both are excluded by Lemma 3.5.

It remains to exclude $OOEOOE$. Let this word be a cycle word at some start, and rotate to a cycle minimum $m \geq 2$. The three alignments of the two even letters are $OOEOOE$, $OEOOEO$, and $EOOEOO$. The last starts even, so it cannot be a minimum, by Theorem 3.2(ii). The middle starts OE: the first image is even and strictly below m^2, whereas every even state after a cycle minimum is at least m^2, as proved in Theorem 3.2(iii). Thus the minimum orientation is $OOEOOE$. In particular m is odd, so $m \geq 3$. The prefix OOE must be realized. If $m = 3$, then $3 \to 5 \to 11$ and 11 is odd, so OOE is not realized. Hence $m \geq 5$. Write $y = J^3(m)$ for the state after OOE. Then $y \geq m$ by minimality, so $y \geq 5$. The suffix OO is realized at y, and Lemma 3.4(i) gives $J^2(y) \geq (y+1)^2 \geq (m+1)^2$. The last letter is even, so Lemma 3.4(iv) gives $J^2(y) < (m+1)^2$. □

Lemma 3.4(v) excludes every odd-run-then-even word O^aE with $a \geq 3$, of any length.

Lemma 3.7 (two length-seven exclusions). Neither $OOOOEOE$ nor $OOOOOEE$ is a cycle word at any $n \geq 2$.

Proof. First, if $n \geq 14$, then

$$n^{243} > 2^{422}(n+1)^{128}.$$

Indeed $14(n+1) \leq 15n$, so $(n+1)^{128} \leq (15/14)^{128} n^{128}$. The finite comparison $2^{422} 15^{128} < 14^{243}$ then yields $2^{422}(n+1)^{128} < 14^{115} n^{128} \leq n^{243}$.

For $2 \leq n < 14$, neither word is realized: at every such start, some letter fails to match the current parity. The same finite check is the Lean `native_decide` evaluation behind `no_cycle_word_ooooeoe` and `no_cycle_word_oooooee` (Appendix A).

Now suppose $OOOOOEE$ is a cycle word at $n \geq 14$, and write $z = J^5(n)$ for the image after the prefix $OOOOO$. Lemma 3.4(iv) on the last even letter, together with the preceding even letter, gives $z < (n+1)^4$. Lemma 3.3 on $OOOOO$ gives $n^{243} \leq 2^{422} z^{32}$, so $n^{243} < 2^{422}(n+1)^{128}$, contradicting the tail inequality.

Finally suppose $OOOOEOE$ is a cycle word at $n \geq 14$. Write $z_4 = J^4(n)$ and $y = J(z_4) = \lfloor \sqrt{z_4} \rfloor$, so $z_4 < (y+1)^2$. Lemma 3.3 on $OOOO$ gives $n^{81} \leq 2^{130} z_4^{16} < 2^{130}(y+1)^{32}$. Cubing yields $n^{243} < 2^{390}(y+1)^{96}$. The last letters OE give the odd-cell bound $y^3 < (n+1)^4$. Write $A = n+1 \geq 15$. The same comparison $(y+1)^3 < 2A^4$ as in Lemma 3.5 holds at this smaller scale. Raising to the thirty-second power gives $(y+1)^{96} < 2^{32}(n+1)^{128}$. Combining with the cubed lower envelope produces again $n^{243} < 2^{422}(n+1)^{128}$. □

Theorem 3.8 (small-cycle census through length seven). No word of length at most seven is a cycle word at any $n \geq 2$. Equivalently, a nontrivial Juggler cycle, if one exists, has period at least eight.

Proof. The reduction of Theorem 3.6 applies at every length: an all-odd word cannot return, and every mixed cycle word rotates to an even-terminating cycle word based at a cycle state $m \geq 2$. Lengths at most six are Theorem 3.6. It remains to exclude even-terminating cycle words of length seven.

By Theorem 3.2(i) a cycle word is formally expanding. The even-terminating expanding length-seven words are exactly O^6E, EO^5E, OEO^4E, $OOEO^3E$, O^3EO^2E, O^4EOE, and O^5EE. Lemma 3.4(v) excludes O^6E. The word EO^5E rotates one step onto O^5EE, and OEO^4E starts OE, so it cannot be a cycle minimum (Theorem 3.2(iii)) and rotates two steps onto O^4EOE; both leftovers are excluded by Lemma 3.7.

It remains to exclude $OOEO^3E$ and O^3EO^2E. Rotate either word to a cycle minimum $m \geq 2$. For $OOEO^3E$ the minimum orientation retains the internal even letter followed by the suffix OOO. Then $m \geq 3$, the prefix through that even letter is realized, and Lemma 3.4(ii) at threshold 3 contradicts the last even cell. For O^3EO^2E the same bootstrap applies with suffix OO and threshold 5, once $m = 3$ is removed: at $m = 3$ the state after OOO is even, so the next even letter is not realized. □

Theorems 3.6 and 3.8 are supporting period statements. The structural claim is the even-count exclusion of Theorem 3.22. The same cells organise leftover words by even-count. Write

$$e_a = 2(3^a - 2^a)$$

for $a \geq 0$.

Lemma 3.9 (trailing even run). If a cycle word based at n ends with $r \geq 1$ even letters, the state immediately before that run is strictly less than $(n+1)^{2^r}$.

Proof. The case $r = 1$ is Lemma 3.4(iv). Suppose the claim holds for some $r \geq 1$, and let vE^{r+1} be a cycle word at n. Write $z = J^{|v|}(n)$. The inductive hypothesis applied to the suffix E^r after the first of those even letters gives $J(z) < (n+1)^{2^r}$. The state z is even, so $z < (J(z)+1)^2 \leq \left((n+1)^{2^r}\right)^2 = (n+1)^{2^{r+1}}$. □

Lemma 3.10 (odd-run lower envelope). If $n \geq 1$ realizes O^a, then

$$n^{3^a} \leq 2^{e_a} J^a(n)^{2^a}.$$

Proof. Lemma 3.3 supplies a multiplicative constant C_v along any realized word, with $C_\varepsilon = 1$, $C \mapsto C \cdot 4^{2^j}$ on an even letter, and $C \mapsto C^3 \cdot 4^{2^j}$ on an odd letter, at step j. On the pure odd word O^a every letter is odd, so $C_{O^{a+1}} = C_{O^a}^3 \cdot 4^{2^a}$. Writing $C_{O^a} = 2^{e_a}$ with $e_0 = 0$ yields the recurrence $e_{a+1} = 3e_a + 2^{a+1}$, because $4^{2^a} = 2^{2^{a+1}}$. The closed form $e_a = 2(3^a - 2^a)$ satisfies the recurrence and the initial value. □

Lemma 3.11 (seven-odd window). No integer n with $2 \le n < 256$ realizes the word O^7.

Proof. This is a table of 254 seven-step evaluations: at every such start, some letter fails to match the current parity. The same finite check is the Lean `native_decide` evaluation behind `no_follows_seven_odds_of_lt256` (Appendix A). □

Theorem 3.12 (two-even leftover families). Let $k \ge 6$ and $n \ge 2$. Neither $O^{k-2}EE$ nor $O^{k-3}EOE$ is a cycle word at n.

Proof. Appendix D.

A cycle word w at n is *minimum-based* when n is a cycle minimum: $J^j(n) \ge n$ for every $0 \le j < |w|$. The remainder after a proper prefix of a cycle word need not itself be a cycle word at the prefix endpoint. The next statement therefore transports the tail inequality of Theorem 3.12, not the cycle-word exclusion at a later start.

Theorem 3.13 (first-even transport). Let $n \ge 2$. No minimum-based cycle word at n has the form O^aEO^bEE with $a \ge 2$ and $b \ge 4$, or the form O^aEO^bEOE with $a \ge 2$ and $b \ge 3$.

Proof. Appendix D.

The hypothesis that the start is a cycle minimum is essential. If $y < n$, the leftover cell is measured against a larger start and need not contradict the tail at y. In particular, Theorem 3.13 does not assert that those words fail to be cycle words at a non-minimum start. That upgrade is Theorem 3.21.

Theorem 3.14 (three trailing evens). Let $a \ge 6$ and $n \ge 2$. The word O^aEEE is not a cycle word at n.

Proof. Appendix D.

Theorem 3.15 (mixed bunched family $EOEE$). Let $a \ge 5$ and $n \ge 2$. The word O^aEOEE is not a cycle word at n.

Proof. Appendix D.

Theorem 3.16 (mixed bunched family $EOOEE$). Let $a \ge 4$ and $n \ge 2$. The word O^aEOOEE is not a cycle word at n.

Proof. Appendix D.

Theorem 3.17 (mixed bunched family $EOOOEE$). Let $a \ge 3$ and $n \ge 2$. The word $O^aEOOOEE$ is not a cycle word at n.

Proof. Appendix D.

Theorem 3.18 (mixed bunched family $EEOE$). Let $a \ge 5$ and $n \ge 2$. The word O^aEEOE is not a cycle word at n.

Proof. Appendix D.

Theorem 3.19 (mixed bunched family $EOEOE$). Let $a \ge 4$ and $n \ge 2$. The word O^aEOEOE is not a cycle word at n.

Proof. Appendix D.

Theorem 3.20 (mixed bunched family $EOOEOE$). Let $a \ge 3$ and $n \ge 2$. The word $O^aEOOEOE$ is not a cycle word at n.

Proof. Appendix D.

Theorem 3.21 (gapped leftovers as cycle words). Let $n \ge 2$. No cycle word at n has the form O^aEO^bEE with $a \ge 2$ and $b \ge 4$, or the form O^aEO^bEOE with $a \ge 2$ and $b \ge 3$.

Proof. Appendix D.

Theorems 3.12–3.21 assemble into an even-count exclusion: no cycle word has fewer than four even letters, so a nontrivial cycle has period at least eleven (Theorem 3.22). Section 4 excludes later periods by financing.

Every integer $2 \le n < 12$ reaches 1, so a cycle minimum is at least 12. Write e for the number of even letters. Lemma 3.21b is the canonical run form.

Lemma 3.21a (classification). Every minimum-based cycle word with at most three even letters belongs to one of the families excluded by Lemma 3.4(v) and Theorems 3.12–3.21.

Proof. Lemma 3.21b puts the word in the form $O^aEO^bEO^cE$ with $a \ge 2$ and unused runs empty. If $e = 0$, the word is all-odd. If $e = 1$, it is O^aE. If $e = 2$, a last run $c \ge 2$ is the internal-even bootstrap of Lemma 3.4, and the remaining shapes are the two-even families of Theorem 3.12. If $e = 3$, again $c \ge 2$ is the bootstrap, while $c \in \{0, 1\}$ is either a gapped leftover (Theorem 3.21) or one of the seven bunched families (Theorems 3.14–3.20). ☐

e	remaining minimum-based forms	elimination
0	all-odd	cannot return (Theorem 3.6)
1	O^aE	next-square; Lemma 3.4(v)
2	O^aEE, O^aEOE, last run ≥ 2	Theorem 3.12; Lemma 3.4
3	bunched families and gapped leftovers	Theorems 3.14–3.21
≥ 4	not excluded by even-count	finance (Section 4)

Theorem 3.22 (even-count). No word with fewer than four even letters is a cycle word at any $n \ge 2$. Equivalently, a nontrivial cycle word has at least four even letters.

Proof. Every cycle word has a minimum-based rotation, with the same even-count. Lemma 3.21b puts that orientation in canonical run form; Lemma 3.21a names the families. In detail:

If $e = 0$, the word is all-odd and cannot return, as in Theorem 3.6.

If $e = 1$, the word is O^aE with $a \ge 2$. The case $a = 2$ is OOE, excluded in Theorem 3.6. The cases $a \ge 3$ are Lemma 3.4(v).

If $e = 2$, the word is O^aEO^cE with $a \ge 2$. A last odd-run $c \ge 2$ is an internal even letter followed by OO or OOO. Lemma 3.4 at the cycle minimum ($n \ge 12$) contradicts the last-even cell. The remaining shapes are O^aEE and O^aEOE. Expansion forces $a \ge 4$ and $a \ge 3$ respectively, so both are Theorem 3.12.

If $e = 3$, the word is $O^aEO^bEO^cE$ with $a \ge 2$. Again $c \ge 2$ is the internal-even bootstrap. The remaining last runs are $c = 0$ and $c = 1$. For $c = 0$ and $b \ge 4$ the word is a gapped leftover O^aEO^bEE, excluded by Theorem 3.21. For $c = 0$ and $b \le 3$ the word is one of O^aEEE, O^aEOEE, O^aEOOEE, $O^aEOOOEE$, excluded by Theorems 3.14–3.17 once expansion supplies the stated lower bounds on a. For $c = 1$ and $b \ge 3$ the word is a gapped leftover O^aEO^bEOE, excluded by Theorem 3.21. For $c = 1$ and $b \le 2$ the word is one of O^aEEOE, O^aEOEOE, $O^aEOOEOE$, excluded by Theorems 3.18–3.20.

Thus $e \le 3$ is impossible. ☐

Corollary 3.23. A nontrivial cycle, if one exists, has period at least eleven.

Proof. Theorem 3.22 gives four even letters, hence $o \le L - 4$. Formal expansion is $2^L < 3^o \le 3^{L-4}$. The comparison $2^L < 3^{L-4}$ first holds at $L = 11$. ☐

In particular there is no cycle of length eight, nine, or ten.

4. Cycle finance

A cycle word is formally expanding (Theorem 3.2), yet the orbit returns exactly. The multiplicative surplus $3^o - 2^L$ must be financed by the floor remainders, which are relatively $O(1/x)$ in logarithms. The resulting bound on the cycle minimum excludes every period that is not admissible for the inequality, once a uniform logarithmic floor-error bound is available above a verified descent floor.

The identity is unrolled along a circular word. After rotation to a cycle minimum that word is a necklace of odd-run excursions. The geometry below records that itinerary. It introduces no new theorem: every named constraint is Theorem 3.2, Lemma 3.4, Lemma 3.21b, or the last-even cell.

The excursion necklace

Write n for a cycle minimum and

$$w = O^{a_1} E O^{a_2} E \cdots O^{a_e} E$$

for a minimum-based orientation (Lemma 3.21b), so $a_1 \geq 2$, $\sum_i a_i = o$, and $e = L - o$. The odd landings are the *valleys* v_i and the even state just before each final E is the *peak* p_i:

$$v_0 = n, \qquad p_i = J^{a_{i+1}}(v_i), \qquad v_{i+1} = J(p_i) = \lfloor \sqrt{p_i} \rfloor \quad (0 \leq i < e),$$

with $v_e = n$. The itinerary is

$$n \xrightarrow{OO} \text{first high region} \xrightarrow{E} v_1 \xrightarrow{O^{a_2} E} \cdots \xrightarrow{O^{a_{e-1}} E} v_{e-1} \xrightarrow{O^{a_e} E} n.$$

Two meanings of *entry* must not be conflated. *Cycle entry* is the distinguished cut that places the minimum at the start of the word. *Dynamical entry* is the last even step into n. The second is a genuine boundary condition; the first is a choice of origin.

Cycle minimum and the forced lift The minimum is odd, so the word cannot begin with E or OE, and it cannot end with an odd letter (Theorem 3.2). The first two letters are therefore OO:

$$n \xrightarrow{O} y \xrightarrow{O} z.$$

For $n \geq 5$ one has $z = J^2(n) \geq (n+1)^2$ (Lemma 3.4(i)). In particular OOE cannot be a cycle word (`no_cycle_word_ooe`). On a cycle minimum the first even residual overshoots the entry cell: the first peak satisfies $p_0 \geq (n+1)^2$. That is the opposite of the last-peak condition below. Even $J^2(n)$ may continue with E; odd $J^2(n)$ continues with O. Either way the minimum-based prefix is still OO.

Excursions, valleys, and peaks An ordinary excursion is one block $O^a E$. In the itinerary semantics that word is `oddEvenBlock a 1`. Write

$$\mu(a) = \frac{3^a}{2^{a+1}}$$

for the ideal (floor-free) exponent of the block. The block is formally expanding if and only if $\mu(a) > 1$, equivalently $2^{a+1} < 3^a$. Thus OE contracts ($\mu(1) = 3/4$) and OOE expands ($\mu(2) = 9/8$). This is a reparameterization of the word envelope of Theorem 2.2, not a transition law on pairs (a_i, a_{i+1}). Lemma 3.4(v) forbids $O^a E$ as a *cycle word* for $a \geq 3$; it does not forbid an internal block of that shape.

The orbit is then the wave

$$v_0 \to p_0 \to v_1 \to p_1 \to \cdots \to v_{e-1} \to p_{e-1} \to v_0.$$

On a cycle minimum every even state is already at least n^2 (Theorem 3.2(iii)). Valleys dominate a logarithmic defect sum because $1/(x \log x)$ is largest there; peaks are huge and cheap. That is why the run-type packing of Theorem 4.7 charges `OOE`-scale valleys, `OE`-scale valleys, and evens separately. The packed comparison is an extremal charge of the same sum, not a uniqueness theorem for the actual word. Expanding blocks can climb and a later OE can drop without crossing the anchor: four consecutive expanding blocks occur already at the certified start 1999 recorded in Section 6.

Closure and the entry cell The valley sequence is circular. The last letter is E, so the last peak occupies the last-even cell of Lemma 3.4(iv):

$$n^2 \le p_{e-1} < (n+1)^2.$$

The minimum is odd, so $p_{e-1} \ne n^2$ (`cycle_last_even_ne_odd_sq`) and p_{e-1} is even. Thus

$$n^2 + 1 \le p_{e-1} < (n+1)^2, \qquad p_{e-1} \text{ even}.$$

An ordinary excursion needs only $v_i \ge n$. The last excursion must hit this cell and land on n:

$$v_{e-1} \xrightarrow{O^{a_e}} p_{e-1} \xrightarrow{E} n.$$

The first peak overshoots the same cell; the last peak lands in it. Those are different even states.

Once the orbit returns to n, the prefix OO is forced again. A genuine cycle is a closed necklace of excursions whose first block starts OO, whose last peak lies in the entry cell, and whose entire orbit stays at least n. The global constraints already proved are

$$\sum_i a_i = o, \qquad e = L - o, \qquad 2^L < 3^o,$$

together with periodicity $\prod_{i=0}^{e-1} v_{i+1}/v_i = 1$.

What remains Finance (Theorem 4.4) sits around the necklace: the surplus $3^o - 2^L$ must be paid by floor losses, and at the verified floor $N_0 = 10^6$ this forces $L \ge 25781$. The run-type packing of Theorem 4.7 is a refinement of the same defect sum along the valleys and peaks just named. Subsequent refinements of one component of the necklace recovered Theorem 4.7 or closed. They are not claims of this note.

The pieces are understood separately: the forced lift at n, the blocks $O^{a_i}E$, and the last-even landing. What is not proved is a link strong enough to exclude the leftover lengths,

$$\text{minimum geometry} + \text{necklace of excursions} + \text{entry cell} \;\Rightarrow\; \perp.$$

That is a formulation of the remaining cycle problem, not a theorem. A genuine lower bound on the number of odd runs would feed Theorem 4.7; none is proved here.

Throughout this section write $L = |w|$ and $o = \#O(w)$ for a cycle word w based at a cycle minimum $n \ge 2$. Natural logarithms are written log. The unique one-step fibres of Section 3 give, for every state $x \ge 1$ with image $y = J(x)$,

$$y^2 \le x^e < (y+1)^2, \qquad e = \begin{cases} 1, & x \text{ even}, \\ 3, & x \text{ odd}. \end{cases}$$

Lemma 4.1 (dyadic-cell logarithm). If $z, y \ge 1$ and $z < (y+1)^2$, then $\log z \le 2\log y + 2/y$.

Proof. The hypothesis gives $\log z \le 2\log(y+1)$. The inequality $\log(1+u) \le u$ for $u > 0$ yields $\log(y+1) = \log y + \log(1 + 1/y) \le \log y + 1/y$. □

Lemma 4.2 (one-step bounds). Let $x \ge 2$ and $y = J(x)$. If x is even, then $\log x \le 2\log y + 2/y$. If x is odd, then $3\log x \le 2\log y + 2/y$.

Proof. The even cell is $y^2 \le x < (y+1)^2$. The cell logarithm lemma on $z = x$ is the first claim. The odd cell is $y^2 \le x^3 < (y+1)^2$. The same lemma on $z = x^3$ gives $\log(x^3) \le 2\log y + 2/y$. □

On a cycle minimum every proper prefix is non-contracting: a prefix with $3^{\#O} < 2^k$ would satisfy $J^k(n) < n$ by Corollary 2.3, contradicting minimality. Thus $2^k \le 3^{o_k}$ for every prefix of length k, where o_k is the odd count of that prefix.

Lemma 4.3 (unrolled envelope). Write $x_k = J^k(n)$ and o_k for the odd count of the length-k prefix. For every $0 \le k \le L$,

$$3^{o_k} \log n \le 2^k \log x_k + \frac{k\, 3^{o_k}}{n}.$$

Proof. The case $k = 0$ is an equality. Suppose the claim holds at $k < L$, and write $x = x_k$ and $y = x_{k+1}$. Minimality gives $n \le y$, and the prefix law gives $2^{k+1} \le 3^{o_{k+1}}$, hence

$$\frac{2^{k+1}}{y} \le \frac{3^{o_{k+1}}}{n}.$$

If the next letter is even, then $o_{k+1} = o_k$ and the one-step bound gives $\log x \le 2 \log y + 2/y$. Multiply by 2^k and add the inductive remainder:

$$3^{o_k} \log n \le 2^{k+1} \log y + \frac{2^{k+1}}{y} + \frac{k\, 3^{o_k}}{n} \le 2^{k+1} \log y + \frac{(k+1)\, 3^{o_k}}{n}.$$

If the next letter is odd, then $o_{k+1} = o_k + 1$. Multiply the inductive bound by 3 and apply the one-step bound in the form $3 \log x \le 2 \log y + 2/y$:

$$3^{o_k+1} \log n \le 2^{k+1} \log y + \frac{2^{k+1}}{y} + \frac{k\, 3^{o_k+1}}{n} \le 2^{k+1} \log y + \frac{(k+1)\, 3^{o_k+1}}{n}.$$

This is the claim at $k + 1$. □

Theorem 4.4 (finance). Let w be a cycle word of length L with o odd letters, based at a cycle minimum $n \ge 2$. Then

$$n \log n \cdot (3^o - 2^L) \le L \cdot 3^o.$$

Proof. Apply Lemma 4.3 at $k = L$. Periodicity gives $x_L = n$ and $o_L = o$, so

$$3^o \log n \le 2^L \log n + \frac{L \cdot 3^o}{n}.$$

The word is formally expanding, so $3^o > 2^L$. Rearranging and multiplying by n is the claim. □

This is the conceptual centre of the note. Exact cells give a one-step logarithmic defect; cycle minimality lets the defect be unrolled against the cycle minimum; the formal surplus $3^o - 2^L$ must then be paid by a finite accumulated budget. Ideal dynamics expands and exact dynamics returns, so the floor errors finance the expansion. The inequality $\log(1 + u) \le u$ is the only analytic input; the content is that interaction. The Lean form is exactly Theorem 4.4 (constant 1): `cycleMin_finance`.

Hierarchy of forms. These three layers must not be conflated.

1. *Theorem 4.4* is the conceptual sharp inequality. Constant 1:

   $$n \log n \cdot (3^o - 2^L) \le L \cdot 3^o,$$

   equivalently $\theta \le L/(n \log n)$ with $\theta = 1 - 2^L/3^o$. It charges every cell defect at the cycle minimum. Lean: `cycleMin_finance`.

2. *Corollary 4.4c* (below) is the same cell-log unroll with remainders kept as $1/x_{i+1}$. It is the strongest proved form of those defects. Lean: `cycleMin_finance_inv_sum`.

3. *Corollary 4.5* is the convenient length-only statewise bound: a verified descent floor N_0 plus a parity charge of the defect sum produces a per-length threshold $n_{\max}(L)$ and excludes every L with $n_{\max}(L) \le N_0$.

4. *Theorem 4.6* is the numerical certification of Corollary 4.5 at $N_0 = 10^6$. The certified identity is the conservative relative-defect form

   $$1 - \frac{2^L}{3^o} \le \frac{6}{5} \sum_{i=1}^{L} \frac{1}{x_i \log x_i}.$$

> The factor 6/5 is a convenient uniform majorant of $-\log(1-\delta)/\delta$ on $[0,1/6]$. It is not Theorem 4.4, and the headline cutoff 25781 is not an artifact of that majorant. The identity itself is Lean for any cycle minimum $n \geq 400$ (`cycleMin_defect_finance`, `DefectFinance.lean`); the per-length numeric table stays a verified computation.

The relative-defect unroll is a parallel identity, not Theorem 4.4 multiplied by 6/5. Charging every state at the cycle minimum in that identity recovers the coarser comparison $\theta \leq (6/5)L/(n\log n)$, which is Theorem 4.4 with coefficient 6/5. At this floor that uniform charge excludes only through length 1053. The computational table uses a stricter length-only parity charge of the same identity.

Corollary 4.4c (inv-sum). Let w be a cycle word of length L with o odd letters, based at a cycle minimum $n \geq 2$. Write $x_i = J^i(n)$. Then

$$(3^o - 2^L)\log n \leq 3^o \sum_{i=1}^{L} \frac{1}{x_i},$$

equivalently $\theta \leq (\sum_i 1/x_i)/\log n$.

Proof. The same induction as Lemma 4.3, keeping each cell defect as $2^{k+1}/x_{k+1}$ instead of replacing it by $3^{o_{k+1}}/n$. At $k = L$ one has $x_L = n$. Lean: `cycleMin_log_envelope_inv`, `cycleMin_finance_inv_sum`. □

The computational table of Theorem 4.6 uses a weaker per-step bound, valid on every cycle because every start below 12 reaches 1 (so every cycle state is at least 12). For a state x with image $y = J(x) \geq 12$, the relative defect $\delta = (x^e - y^2)/x^e$ satisfies $\delta \leq 2/y \leq 1/6$. Writing $\varepsilon = -\frac{1}{2}\log(1-\delta)$ and using $-\log(1-\delta) \leq \frac{6}{5}\delta$ on $[0,1/6]$ gives $\varepsilon \leq (6/5)/y$. Unrolling $t_{i+1} = (e_i/2)\,t_i - \varepsilon_i$ around the cycle then yields the conservative identity displayed in item 4 of the hierarchy. This chain is Lean end to end (`DefectFinance.lean`): the per-step losses in image form (`log_floorPower_even_ge_sub`, `log_floorPower_odd_ge_sub`, from $-\log(1-\delta) \leq \frac{6}{5}\delta$ on $[0,1/6]$, `neg_log_one_sub_le_sixth`), the amplification priced by the upper invariant $\log x_k \leq w_k \log n$ (`cycleMin_log_le_weight`), and the charged unroll (`cycleMin_charge_prefix`), closed at $x_L = n$.

On a minimum-based cycle of length L with o odd letters and $e = L - o$ even letters, the last letter is even, so $e \geq 1$ and the number of odd-run starts is at most e. Every even state satisfies $x \geq n^2$. Every odd state preceded by an odd state satisfies $x \geq t = \lfloor n^{3/2} \rfloor$. Therefore

$$\sum_{i=1}^{L} \frac{1}{x_i \log x_i} \leq \frac{e}{n\log n} + \frac{o-e}{t\log t} + \frac{e}{2n^2\log n},$$

and

$$n\log n\left(1 - \frac{2^L}{3^o}\right) \leq \frac{6}{5}\left(e + (o-e)\frac{n\log n}{t\log t} + \frac{e}{2n}\right).$$

The optimal uniform coefficient on $[0,1/6]$ is $6\log(6/5) \approx 1.093$. Replacing 6/5 by that constant, or even by 1, on the same parity charge does not change the first surviving length: one still has $n_{\max}(25780) \leq 10^6 < n_{\max}(25781)$. The published table and the count 141 keep the proved coefficient 6/5. The cutoff 25781 is therefore not an artifact of an avoidable loss in the majorant.

Lemma 4.4b (odd-count monotonicity). Write $\theta(o) = 1 - 2^L/3^o$ and

$$R(o) = e + (o-e)\alpha + \frac{e}{2n}, \qquad e = L - o, \qquad \alpha = \frac{n\log n}{t\log t}.$$

The certified parity comparison is $n\log n \cdot \theta(o) \leq (6/5)R(o)$. For every $n \geq 12$ one has $\alpha < 1/2$, hence

$$R(o+1) - R(o) = 2\alpha - 1 - \frac{1}{2n} < 0.$$

Also $\theta(o+1) > \theta(o)$. Therefore if the comparison fails at some admissible o, it fails at every larger odd count. Equivalently, the largest n at which the comparison can hold occurs at the least admissible odd count $o_{\min}(L) = \min\{o : 3^o > 2^L\}$. The same monotonicity holds for any positive coefficient in place of 6/5, and for the constant-1 comparison of Theorem 4.4.

Proof. The difference $R(o+1)-R(o)$ is the displayed coefficient of o. For $n \geq 12$ one has $t = \lfloor n^{3/2} \rfloor \geq n^{3/2} - 1 > 2n$, hence $\alpha < (n \log n)/(2n \log(2n)) = (\log n)/(2 \log(2n)) < 1/2$. The map $o \mapsto \theta(o)$ is strictly increasing on integers o with $3^o > 2^L$. If $n \log n \cdot \theta(o) > (6/5)R(o)$, then $n \log n \cdot \theta(o+1) > (6/5)R(o) > (6/5)R(o+1)$. $\square$

Define

$$\gamma(L) = \frac{3^{o_{\min}(L)}}{2^L} - 1,$$

and write $n_{\max}(L)$ for the largest integer n at which the displayed parity inequality can still hold at $o = o_{\min}(L)$. The coarser comparison $n \log n \leq (6/5)L \cdot 3^{o_{\min}}/(3^{o_{\min}} - 2^L)$ is used only as a check; Theorem 4.6 uses $n_{\max}(L)$ from the parity form. Lemma 4.4b is why that table may be computed at $o_{\min}$ alone.

Proposition 4.4a (finance-survivor algorithm). Fix a verified descent floor N_0. For each integer $1 \leq L \leq 10^5$, compute $o_{\min}(L) = \min\{o : 3^o > 2^L\}$ by exact integer arithmetic and $n_{\max}(L)$ from the parity inequality, and retain L if and only if $n_{\max}(L) > N_0$. The resulting finance-survivor set is

$$\mathcal{E}(N_0) = \{L : 1 \leq L \leq 10^5,\ n_{\max}(L) > N_0\}.$$

The printed instance is $\mathcal{E} = \mathcal{E}(10^6)$, with $|\mathcal{E}| = 141$.

Corollary 4.5. If every integer $2 \leq n \leq N_0$ reaches 1, then no nontrivial cycle of length L exists whenever $n_{\max}(L) \leq N_0$.

Proof. A periodic state never reaches 1, so every cycle state is at least $N_0 + 1$. The length-only parity charge of the certified relative-defect identity then forces the minimum to satisfy the displayed inequality, hence $n \leq n_{\max}(L)$. This is the convenient statewise bound in the hierarchy after Theorem 4.4; it is not a replacement for that theorem. $\square$

Record values of the parity $n_{\max}$ include $n_{\max}(19) = 133$, $n_{\max}(84) = 2323$, $n_{\max}(569) = 23568$, $n_{\max}(1054) = 788014$, and $n_{\max}(25781) = 26254995$.

Theorem 4.6 (verified computation). Every integer $2 \leq n \leq 10^6$ reaches 1. Consequently:

(A) there is no nontrivial Juggler cycle of length at most 25780;

(B) if a nontrivial cycle has period $L \leq 10^5$, then $L \in \mathcal{E}$, where $|\mathcal{E}| = 141$.

The bound closes continuously through 25780; 25781 is the first integer for which this particular inequality does not exclude a cycle. The coefficient 6/5 is the certification majorant of item 4 in the hierarchy after Theorem 4.4, not the source of the cutoff.

Proof. The descent floor is Proposition 1.3. The gap table computes $o_{\min}(L)$ and $n_{\max}(L)$ for every $1 \leq L \leq 10^5$ by Lemma 4.4b. Corollary 4.5 at $N_0 = 10^6$ excludes every L with $n_{\max}(L) \leq 10^6$. The contiguous excluded prefix is $L \leq 25780$. The complementary set in range is $\mathcal{E}$; checksums are Appendix B. $\square$

Run packing

Theorem 4.7 refines the length-only charge by separating the unique minimum, `OOE`-scale valleys, `OE`-scale valleys, internal odds, and evens. It is a supporting comparison at the same floor; the leftover count $141 \to 99$ does not raise the cutoff. A cycle cannot put an `OE`-start at the minimum (Theorem 3.2), and an n-circuit of k odds and ℓ evens stays at least n only if $3^k \geq 2^{k+\ell}$.

Theorem 4.7 (run-type packing). Let w be a cycle word of length L with $o = o_{\min}(L)$ odd letters and $e = L - o$ even letters, based at a cycle minimum $n \geq 12$. Write v for the least odd integer with $v^3 \geq n^4$, and write $t = \lfloor n^{3/2} \rfloor$. Then $o - e < e$, the largest number of n-scale valleys compatible with the even cap is $o - e$ copies of `OOE`, and the remaining $2e - o$ circuits are `OE` from v. The cycle minimum occurs once, so

$$\sum_{i=1}^{L} \frac{1}{x_i \log x_i} \leq \frac{1}{n \log n} + \frac{o-e-1}{(n+2)\log(n+2)} + \frac{2e-o}{v \log v} + \frac{1}{t \log t} + \frac{o-e-1}{t_+ \log t_+} + \frac{e}{2n^2 \log n},$$

where $t_+ = J(n+2)$. Combined with the 6/5 unroll this is strictly smaller than the parity sum of Corollary 4.5 whenever $2e - o > 0$. Sending the cycle maximum to infinity removes one even term and does not change the valley packing.

Proof. The statement is at $o = o_{\min}(L)$. Lemma 4.4b already excludes every larger odd count from the parity comparison used in Theorem 4.6; the packing is a refinement at that same odd count. Formal expansion at $o_{\min}$ forces $3o < 2L$ on every leftover length in range, equivalently $o - e < e$. The even-cap comparison $3^k \geq 2^{k+\ell}$ is the ideal power envelope; floors only help. An `OE`-start is followed by an even state, so Theorem 3.2 gives $J(v) \geq n^2$ and therefore $v^3 \geq n^4$. Unique visit of the cycle minimum is periodicity. The displayed sum charges one cheap valley at n, the remaining $o - e - 1$ cheap valleys at the next odd integer $n + 2$, the expensive valleys at v, one internal odd at t, the remaining internals at $J(n+2)$, and every even at n^2. Any deeper odd run or any higher valley only decreases the sum. □

Theorem 4.8 (run-type table). At the verified descent floor $N_0 = 10^6$, the packed comparison of Theorem 4.7 excludes the 42 lengths $56347 + 1054k$ for $k = 0, \dots, 41$ and leaves an explicit set $\mathcal{E}_{\text{run}} = \mathcal{E}_{\text{run}}(10^6)$ of 99 lengths. The first survivor remains 25781. In particular, if a nontrivial cycle has period $L \leq 10^5$, then $L \in \mathcal{E}_{\text{run}}$.

Proof. The 141 lengths of Theorem 4.6(B) are tested at $n = 10^6 + 1$ against the packed right-hand side. The comparison is certified on each length: no entry is left uncertain. The 42 excluded lengths are exactly the arithmetic progression named in the statement. The complementary set in that range is $\mathcal{E}_{\text{run}}$. Checksums are Appendix B. The period cutoff remains 25781. □

Arithmetic structure of the finance survivors

The leftover lengths cluster around the continued-fraction approximants of $\log 2/\log 3$. That organizes the table; it does not constrain a hypothetical cycle.

Proposition 4.9 (finance-survivor lattice). Write $v_* = (25781, 16266)$ and $v_{1054} = (1054, 665)$. Then

$$25781 \cdot 665 - 1054 \cdot 16266 = 1,$$

so these two vectors are a unimodular basis of $\mathbb{Z}^2$. Every length in $\mathcal{E}_{\text{run}}$, and every one of the 42 packing deaths of Theorem 4.8, is of the form $(L, o_{\min}) = a\, v_* + b\, v_{1054}$. With the table cap $L \leq 10^5$ the 99 survivors fall in three affine slices of counts 29, 47, and 23. The identification with $\mathcal{E}_{\text{run}}$ is Theorem 4.8. This is a change of coordinates for (L, o), not a relation between hypothetical cycles.

Proof. The determinant identity is integer arithmetic. The generator comparison $3^{665} > 2^{1054} > 3^{664}$ gives $o_{\min}(1054) = 665$. Direct evaluation of $o_{\min}$ on the 141 lengths of Theorem 4.6(B) places each pair $(L, o_{\min})$ on the displayed lattice, with the 42 packing deaths the F_1 continuation $b \geq 29$. □

5. The laboratory instance and the walk-charge envelope

Everything in Sections 2–4 is floor-generic: Corollary 4.5 accepts any verified descent floor. This section first records a second, laboratory-certified instance of the same architecture, then replaces the length-only charge by a coupled exponent-walk charge. The extremal walk is identified exactly as a rotation word, and a Denjoy–Koksma bound over certified Ostrowski blocks produces an envelope valid for *every* length in an explicit window — no per-length census and no dynamic program is needed on that window. The kill table at the laboratory floor then yields the period bound 176251, and the second certified floor raises it to 478245 (Corollary 5.10). Throughout, survivors are finance-survivors in the sense of Section 1; nothing here is a halt theorem.

5.1 The laboratory floor

Proposition 5.1 (laboratory descent floor; certified computational input). Every integer $2 \leq n \leq 26254995$ reaches 1. Precisely: an exact-integer first-passage run records, for each such n, a finite realized word with image strictly below the start; strong induction on that image reaches 1. The run walks the 13127497 odd starts (even starts descend by E) over 106 contiguous chunk records. Three bit-cap seeds

(7110201, 13184021, 13782577) are resolved exactly at a $512 \cdot 10^6$-bit cap with exact integer square roots, the largest intermediate having 298912128 bits (seed 7110201); the maximum first passage is 325 steps (seed 15909091). Certificate hashes are in Appendix B. The floor $26254995 = n_{\max}(25781)$ is the cheapest floor that moves the Theorem 4.6 cutoff.

Theorem 5.2 (raised cutoff; verified computation). At $N_0 = 26254995$ the parity table of Theorem 4.6 excludes every length $L \leq 50507$: any nontrivial Juggler cycle has period at least 50508. The first finance survivor is $L = 50508$ with $n_{\max}(50508) = 162848325$; 19 parity survivors remain through $L = 2 \cdot 10^5$ (run packing leaves 11 at the same cutoff).

Proof. Proposition 5.1 supplies the floor; Corollary 4.5 and the gap table of Proposition 4.4a exclude every L with $n_{\max}(L) \leq 26254995$. The contiguous excluded prefix is $L \leq 50507$; checksums are Appendix B. □

5.2 Transport to a reduced base

The parity and run-pack tables price valleys independently. On a real minimum-based cycle every state is coupled through one closed exponent walk. With a_k the number of odd letters among the first k, set

$$u_k = (1+\mu)a_k - k, \qquad \mu = \log_2(3/2).$$

The defect-free floors give the upper envelope $x_k \leq n^{2^{u_k}}$, and cycle minimality forces $u_k \geq 0$ at every step. The lower envelope requires controlling the accumulated floor losses; that is the transport lemma.

Theorem 5.3 (transport). On a minimum-based cycle with minimum $n \geq 400$, length L, o odd and e even letters, every state satisfies

$$x_k \;\geq\; \left(n\, e^{-D}\right)^{w_k}, \qquad w_k = 2^{u_k}, \qquad D = \frac{1.05\, e}{n} + \frac{0.7\, o}{n^{3/2}}.$$

Proof. Write $\ln x_k \geq w_k \ln n - E_k$. The floor losses give the recursion $E' \leq \frac{3}{2}E + 1.05\, x^{-3/2}$ at an odd letter and $E' \leq \frac{1}{2}E + 1.05\, x^{-1/2}$ at an even letter, using $-\ln(1-t) \leq 1.05\, t$ on $t \leq 0.05$. Unrolling, the amplification from injection j to state k is exactly w_k/w_{j+1}. Odd injections have $x_j \geq n$ and $w_{j+1} \geq \frac{3}{2}$, contributing at most $0.7\, n^{-3/2}$ each; even injections have $x_j \geq n^2$ (Theorem 3.2(iii)) and $w_{j+1} \geq 1$, contributing at most $1.05/n$ each. Hence $E_k \leq w_k D$. □

The transport inequality is Lean end to end in log form: the walk weight $w_k = 2^{u_k} = 3^{a_k}/2^k$ is rational, so the per-step floor losses (`log_floorPower_odd_ge`, `log_floorPower_even_ge`), the exact weight recursion, and the full induction (`cycleMin_transport`, `WalkTransport.lean`) are elementary real arithmetic over the formalized cycle envelopes `cycleMin_iterate_ge`, `cycleMin_even_ge_sq`, and `cycleMin_prefix_pow_le`.

Consequently the cyclic defect sum $\sum_i 1/(x_i \ln x_i)$ is bounded above by the maximum of the walk charge $\sum_k g(u_k)$, $g(u) = 1/(n'^{2^u} 2^u \ln n')$, over all nonnegative closed exponent walks with o up-steps, evaluated at the *reduced base* $n' = n\, e^{-D}$. No free parameter remains; the walk value feeds the 6/5 unroll of Theorem 4.4 exactly as the parity charge did. At the laboratory floor and window lengths, $D \leq 4.6 \cdot 10^{-3}$, so $\ln n' \geq 17.07$.

This consequence is itself Lean end to end (`WalkChargeMax.lean`): writing the charge through the rational weight, $g = 1/(e^{W\nu} W\nu)$ with $W = 2^u = 3^a/2^k$ and $\nu = \ln n'$, exponentiating the transport inequality gives $\sum_k 1/(x_k \ln x_k) \leq \sum_k g(w_k)$ (`cycleMin_defect_le_charge`), and the charge of the realized word is dominated by the hug charge of the same length (`cycleMin_defect_le_hug_charge`, using Theorem 5.4 below), all under the recorded hypothesis $\nu > 0$.

5.3 The adversary is the hug word

Theorem 5.4 (hug exchange). Among nonnegative exponent walks with prescribed (L, o), the *hug word* — take E at every step where $u \geq 1$, else O — is the unique prefix-minimal admissible path. Since g is strictly decreasing in u, the hug word uniquely maximises the walk charge.

Proof. At the first disagreement with any other admissible word, the hug word holds E where the other holds O, because hug takes O only when E is illegal and the two words share the same prefix state. The odd-count gap $\delta_k = a_k(\text{other}) - a_k(\text{hug})$ is a path with steps in $\{-1, 0, +1\}$ that cannot go negative, since $\delta = 0$ restores the same state. Feasible pairs ($(1+\mu)o \geq L$) never strand: when the odd budget is exhausted, $u = \text{surplus} + e_{\text{left}} \geq e_{\text{left}}$, so the remaining evens are legal. Prefix-minimality plus strict monotonicity of g gives unique maximality. The prefix-minimality core is Lean: `hugOdds_le_of_admissible`. □

The analytic half is also Lean, in a strengthened form (`WalkChargeMax.lean`): the charge is antitone in the rational weight (`stateCharge_antitone`, elementary exp monotonicity — no charge integral), so the exact hug word maximises the total charge over *all* admissible exponent walks, not just a fixed (L, o) class (`hug_charge_maximal`). Only the strict within-(L, o) uniqueness of the maximiser remains a human argument.

The statement is about the $u \geq 0$ relaxation, not about realized cycle words. Word-order (Christoffel) prefix-dominance is *false* for this family — the greedy word *OOEO* beats *OOOE* at $(L, o) = (4, 3)$ — so the exchange argument above, not a dominance order, is the correct mechanism.

Realized words do, however, dominate the hug word at the level of odd counts: on any minimum-based cycle word, every length-k prefix carries at least $o_{\min}(k)$ odd letters. This is Lean end to end (`cycleMin_prefix_odds_ge_hug`, `cycleMin_odds_ge_hug`), composing the formalized cycle prefix envelope $2^k \leq 3^{a_k}$ (`cycleMin_prefix_pow_le`) with the hug minimality `hugOdds_least`. It is exactly the sense in which the hug word is the cheapest adversary any hypothetical cycle can present. The survivor-lattice generators of Proposition 4.9 lie on this hug diagonal: $(1054, 665)$, $(25781, 16266)$, and the seed $(50508, 31867)$ all satisfy $o = o_{\min}(L)$ (Lean: `hugOdds_1054`, `hugOdds_lattice_base`, `hugOdds_seed`).

Proposition 5.5 (rotation average). The infinite hug walk is the rotation by $\alpha = \log_2(3/2)$ on $\mathbb{R}/(1+\alpha)\mathbb{Z}$. Unique ergodicity gives the charge-per-letter

$$C_*(n') = \frac{1}{\ln 3} \int_1^3 n'^{\,1-t}\, t^{-2}\, dt \;<\; \frac{1}{\ln 3\, \ln n'}.$$

Proof. Substituting $s = (t-1)\ln n'$ gives $C_* = \frac{1}{\ln 3 \ln n'} \int_0^{2\ln n'} e^{-s}(1 + s/\ln n')^{-2}\, ds$, and the integrand is at most e^{-s}. □

The display and its quantitative sharpening are Lean (`RotationAverage.lean`): the quadratic majorant $t^{-2} \leq 1 - 2(t-1) + 3(t-1)^2$ on $[1, 3]$ — the product with t^2 is $1 + 4(t-1)^3 + 3(t-1)^4$ — turns the Laplace bound into an exact antiderivative evaluation, giving $C_*(n') \leq (1 - \frac{2}{\nu} + \frac{6}{\nu^2})/(\ln 3\, \nu)$ at $\nu = \ln n'$ with no quadrature (`rotation_average_lt`, `rotationAverage_le`, gap form `rotationAverage_gap`). Only the ergodic *identification* of C_* as the infinite-word average — unique ergodicity of the rotation — remains classical prose.

This is the infinite-word average, not a finite-L inequality: on the certified survey the finite leftover charge exceeds C_* by up to $1.57 \cdot 10^{-5}$, so $C_L \leq C_*$ is false. The next two subsections quantify the finite-L error.

5.4 Word identity

Write C_L for the charge-per-letter of the budgeted hug word at $(L, o_{\min}(L))$.

Lemma 5.6 (word identity). For every L, the budgeted hug word at $(L, o_{\min}(L))$ equals the exact rotation L-prefix generated by the integer rule: E at step k if and only if $3^a \geq 2^{k+1}$, where a is the number of odd letters already used. In particular C_L is a Birkhoff average of the rotation.

Proof. The exact rule keeps $u \in [0, 1+\alpha)$, so its L-prefix uses exactly $o_{\min} = \lceil L \log 2/\log 3 \rceil$ odd letters. A first budget-forced divergence between the two words would make the exact prefix use more of one letter than its own total, which is impossible. Lean: `budgetedWord_eq_hugWord`, with the window invariant `hugOdds_pow_ge` / `hugOdds_pow_lt` and minimality `hugOdds_least` (`WalkChargeWords.lean`). □

5.5 Denjoy–Koksma over certified Ostrowski blocks

Theorem 5.7 (block envelope). For the exact rotation prefix of length L at reduced base n',

$$\left|C_L - C_*(n')\right| \;\le\; \frac{2\,s(L)}{L}, \qquad s(L) = \sum_j b_j,$$

for any decomposition $L = \sum_j b_j q_j$ into convergent denominators q_j of $\theta = \log(3/2)/\log 3$.

Proof. The observable $F(u) = n'^{\,1-2^u}/2^u$ decreases on the circle from $F(0) = 1$, so its variation including the wrap jump is < 2. The Denjoy–Koksma inequality (known) bounds the ergodic sum of each block of length q_j within $\mathrm{Var}(F)$ of $q_j C_*$. Summing the $s(L)$ blocks gives the display. □

The denominator list

$$1,\ 2,\ 3,\ 8,\ 19,\ 65,\ 84,\ 485,\ 1054,\ 24727,\ 50508,\ 125743,\ 176251$$

is certified by an interval continued fraction on the big-integer sandwich $2^{17087915} > 3^{10781274}$ and $2^{16785921} < 3^{10590737}$. The sandwich, the resulting real bounds on θ, the shared quotient prefix of the two rational endpoints, and the convergent recurrence are Lean (`theta_sandwich_upper`, `theta_sandwich_lower`, `lower_lt_walkTheta`, `walkTheta_lt_upper`, `cf_lower_prefix`, `cf_upper_prefix`, `theta_convergent_denominators`, `OstrowskiSandwich.lean`); the cylinder-interval bridge from the endpoints to θ itself is classical. The quantitative hypothesis Denjoy–Koksma needs per block is also Lean: with the matching numerators $0, 1, 1, 3, 7, 24, 31, 179, 389, 9126, 18641, 46408, 65049$ (`theta_convergent_numerators`), consecutive pairs are unimodular, all pairs are coprime, every certified convergent satisfies $|\theta - p/q| < 1/q^2$ against the sandwich (`theta_convergents_unimodular`, `theta_convergents_coprime`, `theta_convergent_quality`), and each block's q rotation steps permute the q grid cells (`theta_block_permutations`). Only the classical variation-versus-integral inequality itself remains prose. A Koksma-type bound with constant 1 (that is, $+1/L$) is *false* for this observable; the correct constant is $2s(L)$.

5.6 The window theorem

Theorem 5.8 (uniform window envelope). For every $L \in [50508,\ 301994)$, at the laboratory floor,

$$C_L \;\le\; C_*(n') + \frac{2\,s(L)}{L} \;<\; \frac{1}{\ln 3\,\ln n'}.$$

Proof. Greedy Ostrowski digits obey $b_j \le a_{j+1}$, so with the certified quotients $\theta = [0; 2, 1, 2, 2, 3, 1, 5, 2, 23, 2, 2, 1, ...]$ the digit sum on the window satisfies $s(L) \le 47$. This step is Lean: the digit cap, exact reconstruction $L = \sum_j b_j q_j$, and the digit-sum bound hold for *any* denominator sequence satisfying the convergent recurrence (`ostroDigit_le`, `ostro_sum_eq`, `ostro_digitSum_le`, `OstrowskiNumeration.lean`), and the certified θ instance gives $s(L) \le 47$ for every $L < 301994$ structurally (`theta_digitSum_le`, `greedyDigitSum_le`). The transport deficit keeps $\ln n' \ge 17.07$, hence

$$\frac{2\,s(L)}{L} \le \frac{94}{50508} = 1.87 \cdot 10^{-3} < 0.00514 \le \frac{1}{\ln 3\,\ln n'} - C_*(n'),$$

the last gap from $\int_0^{2\ln n} e^{-s}(1 + s/\ln n)^{-2} ds \le 1 - 2/\ln n + 6/(\ln n)^2$, which is Lean (`rotation_average_le`, gap form `rotationAverage_gap`, `RotationAverage.lean`). Conclude by Lemma 5.6 and Theorem 5.7. □

Scan sharpening (Lean). An exact scan of all 251486 window lengths gives max digit sum $s = 37$ (at $L = 275632$) and a uniform envelope margin of at least 5.48. The digit scan is Lean-verified: for every $L \in [50508, 301994)$ the greedy digits over the certified denominators reconstruct L and sum to at most 37 (`window_digit_scan`, `window_digit_cap`, `window_digit_max`). The window theorem is census-free; the scan only sharpens the constants.

5.7 Kill table and the period bound

The envelope controls the charge; kill decisions are the per-length finance comparison $\theta(L) > \frac{6}{5} B(L) \cdot \text{guard}$ of the Theorem 4.6 architecture, which is Diophantine, not envelope-limited.

The comparison template is itself Lean (`cycleMin_hug_kill_criterion`, `DefectFinance.lean`): every minimum-based cycle at $n \geq 400$ with positive reduced log-base must satisfy $1 - 2^L/3^o \leq \frac{6}{5} \sum_k g(\text{hugWeight}_k)$ at the reduced base, chaining the Lean finance inequality (`cycleMin_defect_finance`, the certified identity of Theorem 4.6) with the defect-to-hug-charge envelope of §5.2. The kill decisions below verify the numeric *failure* of this inequality per length; only that evaluation, and the parity refinement of Corollary 4.5, remain verified computation.

Theorem 5.9 (verified computation; period bound at the laboratory floor). At $N_0 = 26254995$, the walk charge of Theorem 5.3 excludes 18 of the 19 parity survivors of Theorem 5.2 through $L = 2 \cdot 10^5$. Kill margins run from 1.1204 at the seed $L = 50508$ (required improvement over parity 6.87, walk supplies 7.70; deficit $D = 7.4566 \cdot 10^{-4}$), through 1.1195 and 1.1187 at its multiples 101016 and 151524, up to 7.69 at the parity-marginal lengths. The sole walk survivor is $L = 176251$ (margin 0.159; required improvement 48). The combined parity + walk contiguous excluded prefix is 176250: any nontrivial Juggler cycle has period at least 176251.

Proof. Theorem 5.2 leaves the 19 parity survivors. Each is priced by the exact (step, odd-count) lattice dynamic program at the reduced base (Theorem 5.3) and compared against $\theta(L)$ under the 6/5 unroll; substituting the census-free envelope of Theorem 5.8 instead recovers the same 18 kills (margin 1.1196 at $L = 50508$; $L = 176251$ survives at 0.1588). Checksums are Appendix B. □

Calibration. At the base instance $(L, N_0) = (25781, 10^6)$ the walk charge gives margin 0.196 — correctly no kill (required improvement 32.5, walk supplies 6.37). The mechanism did not change between the two floors; the target did.

Corollary 5.10 (verified computation; second laboratory floor). Every integer $2 \leq n \leq 162849448$ reaches 1 (first-passage certificate exactly as in Proposition 5.1: the extension segment walks 68297226 odd starts over 547 contiguous chunk records with exact integer square roots and no failures of any kind; the maximum first passage is 433 steps at seed 78641579, and the largest intermediate has 463362780 bits at seed 92502777; hashes in Appendix B). At $N_0 = 162849448$ the parity table excludes the previous survivor cluster $\{50508, 101016, 151524\}$ outright and leaves 25 survivors through $L = 6 \cdot 10^5$; the walk charge of Theorem 5.3 kills the 15 below 478245 (margins 1.198 at 176251 and 352502, up to 8.44); lengths $L \leq 176250$ stay excluded because both exclusions are monotone in the floor. The combined contiguous excluded prefix is 478244: any nontrivial Juggler cycle has period at least 478245.

The sole survivor is the semiconvergent fan member $478245 = 176251 + 301994$: required improvement over parity 19.46, direct walk margin 0.4334 — the obstruction is the Diophantine quality of $|3^o - 2^L|$ along the fan, not the envelope. The kill table recomputes on commodity hardware in minutes (Appendix B).

Beyond $q_{13} = 301994$ the window theorem needs deeper certified quotients of θ, and killing the remaining near-convergent survivors (starting with the fan member $L = 478245$) is a Diophantine question about $|3^o - 2^L|$; neither is attempted here. The survivors are finance-survivors, not candidate cycles, and nothing in this section is a halt theorem.

6. Limitations and future directions

The result does not imply termination. The remaining finance-survivor lengths are uncontrolled, and existence of a cycle at such a length is open.

A start $n \geq 2$ has a *descent certificate* if there exists a realized finite word w with $J^{|w|}(n) < n$. Even starts realize E; an odd start with even image realizes OE. The complement of those two words is the odd-to-odd class. If every start above 1 has some descent certificate, strong induction yields arrival at 1. That hypothesis is not proved, and a uniform run bound on expanding blocks is unavailable: four consecutive expanding blocks occur already at

$$1999 \xrightarrow{OOE} 5169 \xrightarrow{OOOOEE} 50093 \xrightarrow{OOE} 193753 \xrightarrow{OOE} 887471.$$

The next concrete direction is the number p of odd runs on a minimum-based cycle — equivalently, the number of excursions on the necklace of Section 4. The run form already gives $p \le e$ and, because the first odd run has length at least two, $p \le o - 1$, hence $p \le \min(e, o-1) < 0.3691\,L$ on an expanding word. That is only the trivial ceiling. A genuine lower bound on p, or a peak-height / peak-count tradeoff, would feed Theorem 4.7. Neither is proved here. The remaining gap recorded there is the missing link from the forced lift at the minimum, through the complete necklace, to the entry cell; it is not a halt theorem.

The same pattern — a piecewise power map, integer rounding, and a cycle minimum — produces a defect-financing obstruction. The Juggler-specific content is the interaction of $x^{3/2}$ and $x^{1/2}$. Analogous questions for other piecewise floor-power maps are not taken up here.

Lean names are in Appendix A. The computational certificates are Propositions 1.3 and 5.1.

Appendix A. Lean names

The core mathematical lemmas of Sections 2–4 are mechanized in Lean 4. The names below are the corresponding theorems in `formal/Problems/Juggler/`, imported by `Problems.JugglerPaper`. Selected finite classifications of Section 3 are `native_decide` tables (Appendix D). Theorems 4.6 and 4.8 are independently certified computations. Proposition 4.9's arithmetic is Lean; its identification with $\mathcal{E}_{\mathrm{run}}$ is Theorem 4.8. This is not a claim that the paper as a whole is formally verified.

Text	Lean
itinerary semantics	`follows_iff_word`, `image_eq_iterate`, `image_append`
Theorem 2.1	`image_monotone_of_follows`
Theorem 2.2	`power_bound_word`
Corollary 2.3	`power_bound_contracts`
Theorem 2.4	`global_defect_identity`
Theorem 2.5	`global_defect_eq_zero_iff_localsTight`, `global_defect_eq_zero_implies_monochrome`, `power_bound_eq_iff_extremal`
Theorem 2.6	`global_defect_append`
Corollary 2.7	`image_eq_start_defectRatio`
per-step slack	`one_plus_eta_lt_succ_sq`
Lemma 3.1	`odd_cell_unique`
CycleMin	`CycleMin`
Theorem 3.2	`cycle_word_formally_expanding`, `cycleMin_start_odd`, `cycleMax_start_even`, `cycleMin_not_end_odd`, `square_scale_superquadratic`, `cycleMin_to_even_superquadratic`
Lemma 3.3	`lower_growth_word`
Lemma 3.4	`oo_suffix_threshold`, `ooo_suffix_threshold`, `threshold_inherits_odd_append`, `cycle_last_even_interval`, `cycle_last_even_ne_odd_sq`, `no_cycle_odd_run_append_even`, `no_cycle_word_ooe`
odd-run block	`oddEvenBlock`
Lemma 3.5	`no_cycle_word_oooeoe`, `no_cycle_word_ooooee`

Text	Lean
Theorem 3.6	`no_cycle_word_length_le_six`
Lemma 3.7	`no_cycle_word_ooooeoe`, `no_cycle_word_oooooee`
Theorem 3.8	`no_cycle_word_length_le_seven`, with `no_cycle_word_ooeoooe`, `no_cycle_word_oooeooe`
Lemma 3.9	`cycle_trailing_evens_lt`
Lemma 3.10	`lowerDenom_replicate_odd`, `odd_run_lower_growth`
Lemma 3.11	`no_follows_seven_odds_of_lt256`
Theorem 3.12	`no_cycle_word_two_even_ee`, `no_cycle_word_two_even_eoe`
Theorem 3.13	`no_cycleMin_gapped_three_even_ee`, `no_cycleMin_gapped_three_even_eoe`
Theorem 3.14	`no_cycle_word_three_even_eee`, with `no_cycle_word_oooooeee`
Theorem 3.15	`no_cycle_word_three_even_eoee`
Theorem 3.16	`no_cycle_word_three_even_eooee`
Theorem 3.17	`no_cycle_word_three_even_eoooee`
Theorem 3.18	`no_cycle_word_three_even_eeoe`
Theorem 3.19	`no_cycle_word_three_even_eoeoe`
Theorem 3.20	`no_cycle_word_three_even_eooeoe`
Theorem 3.21	`no_cycle_word_gapped_three_even_ee`, `no_cycle_word_gapped_three_even_eoe`
Theorem 3.22	`no_cycle_word_even_count_le_three`
Corollary 3.23	`cycle_word_length_ge_eleven`
Lemma 3.21b	canonical run form; Theorem 3.2
Lemma 3.21a	the case split of Theorem 3.22
Lemma 4.1	`log_le_two_log_add`
Lemma 4.2	`log_step_even`, `log_step_odd`
Lemma 4.3	`cycleMin_log_envelope`
Theorem 4.4	`cycleMin_finance`
Corollary 4.4c	`cycleMin_log_envelope_inv`, `cycleMin_finance_inv_sum`
Lemma 4.4b	odd-count monotonicity; human proof, not Lean
Theorem 4.6	certified identity `cycleMin_defect_finance`, per-step losses `log_floorPower_even_ge_sub`, `log_floorPower_odd_ge_sub`, invariants `cycleMin_log_le_weight`, `cycleMin_charge_prefix` (`DefectFinance.lean`); the numeric table is verified computation
Theorem 4.7	run-type packing; human proof, not Lean
Theorem 4.8	run-type table; verified computation, not Lean
Proposition 4.9	`run_survivor_unimodular`, `run_survivor_seed_F2`, `run_survivor_seed_F3`, `three_pow_step_gt_two_pow_step`, `runSurvivors_length`
Proposition 5.1	laboratory floor; certified computation, not Lean
Theorem 5.2	raised cutoff; verified computation, not Lean

Text	Lean
Theorem 5.3	transport inequality `cycleMin_transport`, per-step losses `log_floorPower_even_ge`, `log_floorPower_odd_ge` (`WalkTransport.lean`); §5.2 consequence `cycleMin_defect_le_charge`, `cycleMin_defect_le_hug_charge` (`WalkChargeMax.lean`)
Theorem 5.4	combinatorial core `hugOdds_le_of_admissible`; cycle-word domination `cycleMin_prefix_odds_ge_hug`, `cycleMin_odds_ge_hug`; charge maximisation `stateCharge_antitone`, `hug_charge_maximal` (`WalkChargeMax.lean`); strict within-(L, o) uniqueness human
Proposition 5.5	ergodic identification human; Laplace bound Lean: `inv_sq_le_quad`, `rotation_average_le`, `rotation_average_lt`, `rotationAverage_le`, `rotationAverage_lt`, `rotationAverage_gap` (`RotationAverage.lean`)
Lemma 5.6	`budgetedWord_eq_hugWord`, `hugOdds_pow_ge`, `hugOdds_pow_lt`, `hugOdds_pow_gt`, `hugOdds_least`
Theorem 5.7	Denjoy–Koksma's variation inequality (known), not Lean; quotient arithmetic `theta_sandwich_upper`, `theta_sandwich_lower`, `lower_lt_walkTheta`, `walkTheta_lt_upper`, `cf_lower_prefix`, `cf_upper_prefix`, `theta_convergent_denominators`; DK hypotheses `theta_convergent_numerators`, `theta_convergents_unimodular`, `theta_convergents_coprime`, `theta_convergent_quality` ($\lvert\theta - p/q\rvert < 1/q^2$), `theta_block_permutations`
Theorem 5.8	digit cap Lean: general numeration `ostroDigit_le`, `ostro_sum_eq`, `ostro_digitSum_le`, instance `theta_digitSum_le`, `greedyDigitSum_le`; scan `window_digit_scan`, `window_digit_cap`, `window_digit_max`; Denjoy–Koksma comparison human
Theorem 5.9	kill template `cycleMin_hug_kill_criterion` (`DefectFinance.lean`); the per-length kill table is verified computation
Corollary 5.10	second floor and kill table; verified computation, not Lean
short certificates (Section 6)	`even_finiteProgress`, `odd_even_finiteProgress`
no certificate ⇒ odd-to-odd	`no_finiteProgress_implies_odd_odd`
induction to 1	`reachesOne_of_all_finiteProgress`
four-block chain	`four_block_pe_1999`

Appendix B. Admissible lengths

The record lengths of $\gamma(L)$ through 10^5, with the parity 6/5 bound $n_{\max}$ of Section 4, are

L	$o_{\min}$	$n_{\max}$
1	1	3
3	2	7
11	7	25
19	12	133
84	53	2323
569	359	23568
1054	665	788014
25781	16266	26254995
50508	31867	162848325

At the verified descent floor $N_0 = 10^6$ the first seven rows are excluded. The set $\mathcal{E} = \mathcal{E}(10^6)$ is defined by Proposition 4.4a: for each $1 \le L \le 10^5$, compute $o_{\min}(L)$ by exact integer arithmetic and $n_{\max}(L)$ from the parity inequality, and retain L if and only if $n_{\max}(L) > 10^6$. This produces 141 lengths. The first few are

$$25781,\ 26835,\ 27889,\ 28943,\ 29997,$$

and a typical combination is $26835 = 25781 + 1054$. The last is 99561. The complete list is the `lengths` array of `data/research/juggler/cycle_finance/exceptions_parity.json`. The SHA-256 of that array, serialized as a JSON list of integers with no spaces, is `dd71aa1527656ba51cb031bafa5497f7bfdbbc43151ffba2c595793326bf7944`. The SHA-256 of the whole file `exceptions_parity.json` is `6b4eec79295b70cdeb9f7db677b7fd57bcb9bd1b51177e1e74aad7b` The SHA-256 of the first-passage file `floor.json` in the same directory is `5b1ce1eec61301cf5b4f969cd5b58954255194e5c7b2` The parity table is written by `research.juggler_sequence.cycle_finance.write_parity_artifacts`. The first-passage file is regenerated by `python -m research.juggler_sequence.cycle_finance`.

The run-type table of Theorem 4.8 is `data/research/juggler/cycle_finance/budget_opt.json`. The 42 excluded lengths are `killed_by_budget`; their SHA-256, serialized as a JSON list of integers with no spaces, is `9d108776d6dc5dc1ae2594058850463cd2d3995cc57b9811471f08e3b818b90a`. The complementary 99 lengths are $\mathcal{E}_{\mathrm{run}}$. Their SHA-256, in the same serialization, is `9e2098923ccb39933630b116133a3fc2ddaf98ace4eb76dbab9b5a` The first few survivors remain

$$25781,\ 26835,\ 27889,\ 28943,\ 29997;$$

the last survivor is 99477. The three families and the unimodular basis are Proposition 4.9. The finite leftover tables of Section 3 are the Lean `native_decide` evaluations named in Appendix A (`LeftoverEval.lean`, `LeftoverShort.lean`, `LeftoverFamilies.lean`). The lattice arithmetic of Proposition 4.9 is `RunSurvivorLattice.lean`.

The laboratory instance of Section 5 has its own artifacts. The first-passage certificate of Proposition 5.1 is `data/research/juggler/cycle_finance/floor_verify/N26254995/certificate.json`; the SHA-256 of its chunk records is `cbcbb540dd860b775d2a3b4351f7cf779609104d3aaf30c342aed6b87f36c9dc`. The parity survivor list of Theorem 5.2 has SHA-256 `d5946efdfac95c715ea46c81979a0eeaf40ad8a1dc893d161bab84ffa7f2afd9`. The walk-charge kill table of Theorem 5.9 is `data/research/juggler/cycle_walk_charge/survey.json`; the SHA-256 of the walk-alive list is `225d76ad12802a690934d01e2d37b3418865441a6825a015dd883c989d8942ec`. The second-floor certificate of Corollary 5.10 is `data/research/juggler/cycle_finance/floor_verify/N162849448/certi` the SHA-256 of its chunk records is `35d9755ce52a225a161b009d0f3674b24a5d0add16035f09ea83cf3880414ae4`. Its parity survivor scan is `data/research/juggler/cycle_walk_charge/new_floor_parity_leftovers.json`, and the 15 kill records under `data/research/juggler/cycle_walk_charge/new_floor_kills/` have SHA-256 `148180cbbfba93985b7a1be455fee16db97816f1534b1feca5a4740d475aeda0` (concatenated in length order); the direct non-kill record for the survivor 478245 (margin 0.4334) is stored alongside. The dynamic program of Theorem 5.3 also has a GPU port (`research.juggler_sequence.cycle_walk_charge_gpu`, CuPy, IEEE double precision) that reproduces the stored CPU kill records to relative error below 10^{-13} at roughly $450\times$ speed; the entire kill table of this note recomputes in minutes on one consumer GPU. The probes are `research.juggler_sequence.cycle_walk_charge` (transport and kill table), `cycle_walk_ostrowski` (certified quotient sandwich and block envelope), and `cycle_walk_window` (window scan); artifacts live under `data/research/juggler/cycle_walk_*`.

Appendix C. Exact floor defect

This appendix records the exact identity behind Theorem 2.2. The present note uses only the nonnegativity $\Delta_w(n) \ge 0$. A nontrivial itinerary-dependent lower bound on Δ_w is left for future work.

For a single branch,

$$x^e = J(x)^2 + \rho(x), \qquad e = \begin{cases} 1, & x \text{ even}, \\ 3, & x \text{ odd}, \end{cases}$$

with $0 \le \rho(x) < 2J(x) + 1$. Write $\operatorname{gap}(a, \rho, e) = (a + \rho)^e - a^e$. The *global defect* $\Delta_w(n)$ is the terminal value of the resulting power-gap recurrence.

Theorem 2.4 (global defect identity). If w is realized at n and $m = J^{|w|}(n)$, then

$$n^{3^{\#O(w)}} = m^{2^{|w|}} + \Delta_w(n), \qquad \Delta_w(n) \ge 0.$$

Theorem 2.2 is the inequality $\Delta_w(n) \ge 0$.

Proof. Induct on w. The empty word is $n = n + 0$. An even letter substitutes $x = J(x)^2 + \rho(x)$ into the inductive identity and lifts the new remainder through 2^ℓ. An odd letter cubes the identity and then substitutes $x^3 = J(x)^2 + \rho(x)$. Each step adds a nonnegative power-gap. □

For a one-letter illustration, $n = 3$ and $w = O$ give $J(3) = 5$ and $3^3 = 27 = 5^2 + 2$, so $\Delta_O(3) = 2$.

Theorem 2.5 (vanishing). If w is realized at n, the following are equivalent: $\Delta_w(n) = 0$; every local remainder along w vanishes; and $\left(J^{|w|}(n)\right)^{2^{|w|}} = n^{3^{\#O(w)}}$. In that case w is monochrome: either $w = E^k$ and $n = a^{2^k}$ for an even a, or $w = O^k$ and $n = a^{2^k}$ for an odd a. A realized mixed word therefore has $\Delta_w(n) > 0$.

Proof. Theorem 2.4 gives the first and third items. A power-gap vanishes if and only if its addend vanishes, so zero defect forces every remainder to vanish, and conversely. Vanishing remainders preserve parity, hence monochrome itineraries, and unique factorization produces the two power towers. □

Theorem 2.6 (composition). If u is realized at n and v is realized at $m = J^{|u|}(n)$, then $\Delta_{uv}(n)$ is the sum of two power-gaps, one lifting $\Delta_u(n)$ through $3^{\#O(v)}$ and one lifting $\Delta_v(m)$ through $2^{|u|}$. In particular $\Delta_v(m)^{2^{|u|}} \le \Delta_{uv}(n)$, and every local remainder satisfies $\rho_j^{2^j} \le \Delta_w(n)$.

Proof. Apply Theorem 2.4 to u, to v, and to uv, and expand the two power-gaps. □

Composition is polynomial, not additive. The Lean form is `global_defect_append`.

Corollary 2.7 (cycle surplus). If w is a cycle word at n, then $\Delta_w(n) = n^{3^{\#O(w)}} - n^{2^{|w|}}$ exactly.

Proof. Theorem 2.4 with $m = n$. □

A mixed word has strict total defect, but the relative slack of a single letter tends to 0 with the state, so no uniform local tax exists. Theorem 4.4 does not use these identities.

Appendix D. Family exclusions

This appendix records the family-by-family exhaustion used by Lemma 3.21a and Theorem 3.22. Each geometry is killed by a next-square obstruction, by long odd-run growth against a last-even cell, or by a finite exceptional window.

Theorem 3.12 (two-even leftover families). Let $k \ge 6$ and $n \ge 2$. Neither $O^{k-2}EE$ nor $O^{k-3}EOE$ is a cycle word at n.

Proof. * First, if $n \ge 256$, then

$$n^{3^{k-2}} > 2^{e_{k-2}}(n+1)^{2^k}.$$

The case $k = 6$ is the tail inequality of Lemma 3.5. If the display holds at some $k \ge 6$, cubing both sides produces

$$n^{3^{k-1}} > 2^{3e_{k-2}}(n+1)^{3 \cdot 2^k}.$$

The recurrence of Lemma 3.10 gives $e_{k-1} = 3e_{k-2} + 2^{k-1}$, so the desired comparison at length $k+1$ reduces to $2^{2^{k-1}} < (n+1)^{2^k}$. Equivalently $2 < (n+1)^2$, which holds for every $n \geq 256$.

Now suppose $O^{k-2}EE$ is a cycle word at such an n. Write $z = J^{k-2}(n)$. Lemma 3.9 with $r = 2$ gives $z < (n+1)^4$. Lemma 3.10 on the prefix O^{k-2} gives $n^{3^{k-2}} \leq 2^{e_{k-2}} z^{2^{k-2}}$, hence $n^{3^{k-2}} < 2^{e_{k-2}}(n+1)^{2^k}$, contradicting the tail.

Finally suppose $O^{k-3}EOE$ is a cycle word at such an n. Write $z = J^{k-3}(n)$ and $y = \lfloor\sqrt{z}\rfloor$, so $z < (y+1)^2$. Lemma 3.10 on O^{k-3} and cubing produce $n^{3^{k-2}} < 2^{3e_{k-3}}(y+1)^{3\cdot 2^{k-2}}$. The last letters OE give the odd-cell bound $y^3 < (n+1)^4$. The comparison $(y+1)^3 < 2(n+1)^4$ of Lemma 3.5 applies at this scale. Raising it to the power 2^{k-2} and using $e_{k-2} = 3e_{k-3} + 2^{k-2}$ recovers again $n^{3^{k-2}} < 2^{e_{k-2}}(n+1)^{2^k}$.

For $2 \leq n < 256$, the cases $k = 6$ and $k = 7$ are Lemmas 3.5 and 3.7. The remaining short words O^6EE, O^5EOE, and O^6EOE fail to return on the same 254-start window; this is the Lean `native_decide` evaluation behind `no_cycle_word_two_even_ee` and `no_cycle_word_two_even_eoe` (Appendix A). Any longer leftover of either family begins with seven consecutive odd letters, which Lemma 3.11 forbids on this window. □

Theorem 3.13 (first-even transport). Let $n \geq 2$. No minimum-based cycle word at n has the form O^aEO^bEE with $a \geq 2$ and $b \geq 4$, or the form O^aEO^bEOE with $a \geq 2$ and $b \geq 3$.

Proof. * Write $y = J^{a+1}(n)$ for the state after the first even letter. Minimum-basedness gives $y \geq n$. In the first family the remainder after that letter is O^bEE with $b+2 \geq 6$; in the second it is O^bEOE with $b+3 \geq 6$. The trailing-even and last-odd cells of those remainders are measured against the cycle start n. Combined with Lemma 3.10 at the remainder start y, the same algebra as in Theorem 3.12 produces

$$y^{3^{\ell-2}} < 2^{e_{\ell-2}}(n+1)^{2^\ell} \leq 2^{e_{\ell-2}}(y+1)^{2^\ell},$$

where ℓ is the remainder length. If $y \geq 256$, the first paragraph of Theorem 3.12 supplies the opposite inequality at y.

If $y < 256$, then $n \leq y < 256$. A gapped leftover of total length at least 17 has $a \geq 7$ or $b \geq 7$, so either the prefix or the remainder realizes seven consecutive odd letters, contradicting Lemma 3.11. The finitely many short-gap words with $2 \leq a \leq 6$ and $b \leq 6$ fail to be minimum-based cycle words on the window $2 \leq n < 256$; this is the Lean `native_decide` evaluation behind `no_cycleMin_gapped_three_even_ee` and `no_cycleMin_gapped_three_even_eoe` (Appendix A). □

Theorem 3.14 (three trailing evens). Let $a \geq 6$ and $n \geq 2$. The word O^aEEE is not a cycle word at n.

Proof. * Write $z = J^a(n)$. Lemma 3.9 with $r = 3$ gives $z < (n+1)^8$. Lemma 3.10 then yields $n^{3^a} < 2^{e_a}(n+1)^{2^{a+3}}$ on any such cycle word. For $n \geq 128$ the opposite comparison

$$n^{3^a} > 2^{e_a}(n+1)^{2^{a+3}}$$

holds. The case $a = 6$ is $n^{729} > 2^{1330}(n+1)^{512}$. Indeed, for $n \geq 128$ one has $(n+1)^{512} < (129/128)^{512}n^{512} < e^4 n^{512} < 64\, n^{512}$, so the claimed bound reduces to $n^{217} > 2^{1336}$. Since $n \geq 128 = 2^7$, the left side is at least 2^{1519}. Cubing the $a = 6$ comparison produces the general case: the recurrence of Lemma 3.10 reduces the inductive step to $(n+1)^4 > 2$, which holds for every $n \geq 128$.

For $2 \leq n < 128$, the case $a = 6$ is a table of evaluations of $OOOOOOEEE$: at every such start the word fails to return. The same finite check is the Lean `native_decide` evaluation behind `no_cycle_word_ooooooeee` (Appendix A). For $a \geq 7$ the prefix contains seven consecutive odd letters, which Lemma 3.11 forbids on this window. □

Theorem 3.15 (mixed bunched family $EOEE$). Let $a \geq 5$ and $n \geq 2$. The word O^aEOEE is not a cycle word at n.

Proof. * First let $n \geq 4$, and write $z = J^a(n)$, $y = \lfloor\sqrt{z}\rfloor$, and $p = J(y)$. Lemma 3.9 with $r = 2$ after the prefix O^aEO gives $p < (n+1)^4$. The letter after y is odd, so Lemma 3.10 at length one yields $y^3 \leq 4p^2 < 4(n+1)^8$.

For $n \ge 4$ one has $4(n+1)^8 < (n+1)^9$, hence $y < (n+1)^3$. The even cell at z then gives $z < (y+1)^2 \le (n+1)^6$. Combined with Lemma 3.10, any such cycle word would satisfy

$$n^{3^a} < 2^{e_a}(n+1)^{6\cdot 2^a}.$$

For $a = 5$ and $n \ge 314$, the opposite comparison $n^{243} > 2^{422}(n+1)^{192}$ holds. The base instance is the finite inequality $2^{422}315^{192} < 314^{243}$. If the display holds at some $n \ge 1$, the elementary comparison $n(n+2) < (n+1)^2$ upgrades it to the same display at $n+1$. Cubing then produces the comparison at $a+1$, once $(n+1)^6 > 4$. In particular the case $a = 6$ already holds for every $n \ge 16$.

For $2 \le n < 314$ and $a = 5$, and for $2 \le n < 16$ and $a = 6$, the word fails to return; these are the Lean `native_decide` evaluations behind `no_cycle_word_three_even_eoee` (Appendix A). For $a \ge 7$ and $n < 256$, Lemma 3.11 applies. For $a \ge 6$ and $n \ge 16$, the tail of the previous paragraph applies. □

Theorem 3.16 (mixed bunched family $EOOEE$**).** Let $a \ge 4$ and $n \ge 2$. The word O^aEOOEE is not a cycle word at n.

Proof. * First let $n \ge 32$, and write $z = J^a(n)$, $y = \lfloor\sqrt{z}\rfloor$, and p for the image after the prefix O^aEOO. Lemma 3.9 with $r = 2$ gives $p < (n+1)^4$. The two letters after y are odd, so Lemma 3.10 at length two yields $y^9 \le 2^{10}p^4 < 2^{10}(n+1)^{16}$. For $n \ge 32$ one has $2^{10} < (n+1)^2$, hence $y < (n+1)^2$. The even cell at z then gives $z < (y+1)^2 \le (n+1)^4$. Combined with Lemma 3.10, any such cycle word would satisfy

$$n^{3^a} < 2^{e_a}(n+1)^{2^{a+2}}.$$

For $n \ge 256$ and $a \ge 4$, this is the opposite of the shared tail of Theorem 3.12 at length $k = a+2$.

For $2 \le n < 256$, the cases $a = 4, 5, 6$ fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_word_three_even_eooee` (Appendix A). For $a \ge 7$, Lemma 3.11 applies. □

Theorem 3.17 (mixed bunched family $EOOOEE$**).** Let $a \ge 3$ and $n \ge 2$. The word $O^aEOOOEE$ is not a cycle word at n.

Proof. * First let $a \ge 4$ and $n \ge 3$, and write $z = J^a(n)$, $y = \lfloor\sqrt{z}\rfloor$, and p for the image after the prefix O^aEOOO. Lemma 3.9 with $r = 2$ gives $p < (n+1)^4$. The three letters after y are odd, so Lemma 3.10 at length three yields $y^{27} \le 2^{38}p^8 < 2^{38}(n+1)^{32}$. For $n \ge 3$ one has $2^{38} < (n+1)^{22}$, hence $y < (n+1)^2$. The even cell at z then gives $z < (y+1)^2 \le (n+1)^4$. Combined with Lemma 3.10, any such cycle word would satisfy

$$n^{3^a} < 2^{e_a}(n+1)^{2^{a+2}}.$$

For $n \ge 256$ and $a \ge 4$, this is the opposite of the shared tail of Theorem 3.12 at length $k = a+2$, already used in Theorem 3.16.

Now let $a = 3$ and $n \ge 256$. The same two-even cell gives $z < (y+1)^2$, so Lemma 3.10 at the prefix O^3 yields $n^{27} < 2^{38}(y+1)^{16}$. The three-odd envelope on y still gives $y^{27} < 2^{38}(n+1)^{32}$.

If $y < 39$, then $n^{27} < 2^{38}\cdot 39^{16}$. The numerical comparison $2^{38}\cdot 39^{16} < 24^{27}$ contradicts $n \ge 24$.

If $y \ge 39$, the numerical comparison $40^{27} < 2\cdot 39^{27}$ upgrades to $(y+1)^{27} < 2y^{27}$, hence $(y+1)^{27} < 2^{39}(n+1)^{32}$. Cubing the prefix bound three times produces $n^{729} < 2^{1026}(y+1)^{432}$. Raising the successor bound to the sixteenth power produces $(y+1)^{432} < 2^{624}(n+1)^{512}$. Combining these displays yields $n^{729} < 2^{1650}(n+1)^{512}$. The opposite comparison holds at $n = 197$ and persists to every larger start by the elementary comparison $n(n+2) < (n+1)^2$ already used in Theorem 3.15.

For $2 \le n < 256$, the cases $a = 3, 4, 5, 6$ fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_word_three_even_eoooee` (Appendix A). For $a \ge 7$, Lemma 3.11 applies. □

Theorem 3.18 (mixed bunched family $EEOE$**).** Let $a \ge 5$ and $n \ge 2$. The word O^aEEOE is not a cycle word at n.

Proof. * First let $n \ge 4$, and write $z = J^a(n)$ and y for the last odd letter of O^aEEOE. The suffix EOE is a cycle suffix, so the last-odd cube of Theorem 3.15 gives $y^3 < (n+1)^4$. The two letters between z and y are even, so $z < (y+1)^4$. For $n \geq 4$ the successor comparison $(y+1)^3 < 2(n+1)^4$ upgrades this to $z < (n+1)^6$. Combined with Lemma 3.10, any such cycle word would satisfy the same display as Theorem 3.15:

$$n^{3^a} < 2^{e_a}(n+1)^{6\cdot 2^a}.$$

The opposite comparison is therefore the tail of Theorem 3.15: it holds for $a = 5$ and $n \geq 314$, and already for $a = 6$ and $n \geq 16$.

For $2 \leq n < 314$ and $a = 5$, and for $2 \leq n < 16$ and $a = 6$, the word fails to return; these are the Lean `native_decide` evaluations behind `no_cycle_word_three_even_eeoe` (Appendix A). For $a \geq 7$ and $n < 256$, Lemma 3.11 applies. For $a \geq 6$ and $n \geq 16$, the tail of the previous paragraph applies. ☐

Theorem 3.19 (mixed bunched family $EOEOE$**).** Let $a \geq 4$ and $n \geq 2$. The word O^aEOEOE is not a cycle word at n.

Proof. * First let $n \geq 32$, and write $z = J^a(n)$, $w = \lfloor\sqrt{z}\rfloor$, and y for the last odd letter. The suffix EOE again gives $y^3 < (n+1)^4$. The one-odd envelope on w yields $w^3 \leq 4s^2$, where s is the image after O^aEO. The last-odd cell and $n \geq 32$ upgrade this to $w < (n+1)^2$, hence $z < (w+1)^2 \leq (n+1)^4$. Combined with Lemma 3.10, any such cycle word would satisfy

$$n^{3^a} < 2^{e_a}(n+1)^{2^{a+2}}.$$

For $n \geq 256$ and $a \geq 4$, this is the shared tail already used in Theorem 3.16.

For $2 \leq n < 256$, the cases $a = 4, 5, 6$ fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_word_three_even_eoeoe` (Appendix A). For $a \geq 7$, Lemma 3.11 applies. ☐

Theorem 3.20 (mixed bunched family $EOOEOE$**).** Let $a \geq 3$ and $n \geq 2$. The word $O^aEOOEOE$ is not a cycle word at n.

Proof. * First let $a \geq 4$ and $n \geq 4$, and write $z = J^a(n)$, $u = \lfloor\sqrt{z}\rfloor$, and y for the last odd letter. The suffix EOE gives $y^3 < (n+1)^4$, hence $(y+1)^3 < 2(n+1)^4$. The two letters after u are odd, so Lemma 3.10 at length two yields $u^9 \leq 2^{10}s^4 < 2^{10}(y+1)^8$. Cubing that display against the last-odd successor bound produces $u^{27} < 2^{38}(n+1)^{32}$. The same comparison as in Theorem 3.17 then gives $u < (n+1)^2$, hence $z < (u+1)^2 \leq (n+1)^4$. Combined with Lemma 3.10, any such cycle word would satisfy the shared two-even tail of Theorem 3.16.

Now let $a = 3$ and $n \geq 256$. The prefix O^3 against $z < (u+1)^2$ yields $n^{27} < 2^{38}(u+1)^{16}$, and the two-odd plus last-odd geometry of the previous paragraph yields $u^{27} < 2^{38}(n+1)^{32}$. These are the same two displays as in the $a = 3$ case of Theorem 3.17, with u in place of y, and the same small/large split applies.

For $2 \leq n < 256$, the cases $a = 3, 4, 5, 6$ fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_word_three_even_eooeoe` (Appendix A). For $a \geq 7$, Lemma 3.11 applies. ☐

Theorem 3.21 (gapped leftovers as cycle words). Let $n \geq 2$. No cycle word at n has the form O^aEO^bEE with $a \geq 2$ and $b \geq 4$, or the form O^aEO^bEOE with $a \geq 2$ and $b \geq 3$.

Proof. * Every cycle word has a minimum-based rotation. It is therefore enough to check that every cyclic shift of either word is an already-excluded cycle-minimum orientation.

Write w for the gapped word. In the first family, $|w| = a + b + 3$. The rotation by $k = 0$ is the original word, excluded as a cycle minimum by Theorem 3.13. The rotation by $k = a+1$ is the bootstrap word O^bEEO^aE. That word has an internal even letter and last gap at least 2. If $a \geq 3$, the last-gap threshold of Lemma 3.4 at OOO and $N = 3$ excludes it. If $a = 2$, the same lemma at OO and $N = 5$ excludes every start $n \geq 5$; the remaining odd start $n = 3$ does not realize four consecutive odd letters, so it cannot follow O^b for $b \geq 4$. The rotation by $k = a + b + 2$ begins with the last even letter of w, which Theorem 3.2 forbids at a cycle minimum. Every other rotation ends with an odd letter, likewise forbidden at a cycle minimum.

In the second family, $|w| = a + b + 4$. The same four classes appear: the original word is Theorem 3.13; the rotation by $k = a + 1$ is O^bEOEO^aE, excluded by the same last-gap thresholds, with the remaining start $n = 3$ and $a = 2$ either failing to realize four odds or, when $b = 3$, reaching 6 after $OOOE$ and then meeting an odd letter; the rotation by $k = a + b + 2$ begins OE, forbidden by Theorem 3.2; and every other rotation ends odd.

The original start need not be a cycle minimum. After rotation the start is a minimum, so the hypothesis $y < n$ that blocked Theorem 3.13 does not arise. $\square$

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